

**1998–2022 Hong Kong (China)
Mathematical Olympiad
Solutions**

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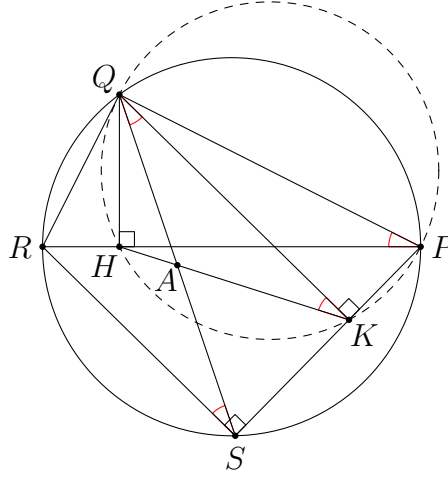
Solutions

Year 1998

1. Let HK meet QS at A . Since $\angle PHQ = \angle PKQ = 90^\circ$, the points P, Q, H, K are concyclic. As both QK and RS are perpendicular to PS , they are parallel. Therefore, we have

$$\angle QKA = \angle QKH = \angle QPH = \angle QPR = \angle QSR = \angle SQK = \angle AQK.$$

Thus, $AK = AQ$. As $\angle QKS = 90^\circ$, A is the midpoint of QS . This means HK bisects QS .



2. Consider the 9 edges on the lateral surface. By the pigeonhole principle, 5 of these edges $PA_1, PA_2, PA_3, PA_4, PA_5$ are of the same colour, say black. WLOG assume A_1, A_2, A_3, A_4, A_5 lie in the this order at the base. Since the base has 9 sides, WLOG assume A_1A_2 is a diagonal. Then A_1A_4 and A_2A_4 are also diagonals. If one of A_1A_2, A_1A_4, A_2A_4 is black, then it forms a monochromatic triangle with P . Otherwise, all of A_1A_2, A_1A_4, A_2A_4 are white, so we still get a monochromatic triangle.
3. (a) Yes. Since $s, t \neq 0$, $s^2 + t^2 \geq 2$. Let p be any prime divisor of $s^2 + t^2$. If $p \mid t$, then $p \mid s$, and so $\gcd(s, t) \geq p$. This shows (s, t) is good. If $p \nmid t$, then there exists $k \in \mathbb{Z}^+$ such that $tk \equiv -s \pmod{p}$. Then we have $p \mid s + kt$. Also, we have

$$t - ks \equiv t - (-t^{-1}s)s \equiv t^{-1}(t^2 + s^2) \equiv 0 \pmod{p}.$$

Thus, $\gcd(s + kt, t - ks) \geq p$. This shows (s, t) is a good pair as we can apply k moves to (s, t) to obtain $(s + kt, t - ks)$.

- (b) Let $\gcd(s, t) = d$. By Bézout's identity, there exist $m, n \in \mathbb{Z}$ such that $ms + nt = d$. We claim that the pair (m, n) is not good.

After k moves, the pair becomes $(m + kt, n - ks)$. If there exists a prime p dividing both $m + kt$ and $n - ks$, then

$$p \mid (m + kt)\frac{s}{d} + (n - ks)\frac{t}{d} = \frac{ms + nt}{d} = 1.$$

This is impossible. Therefore, $\gcd(m + kt, n - ks) = 1$ for any k . This proves our claim.

4. (a) We label the inequality as follows.

$$g(ty + (1 - t)z) \leq tg(y) + (1 - t)g(z) \quad (1)$$

Putting $y = n - 1$ and $z = n + c$ in (1), we obtain

$$g(t(n - 1) + (1 - t)(n + c)) \leq tg(n - 1) + (1 - t)g(n + c).$$

Suppose $c \in [0, 1]$. Replacing t by $\frac{c}{1 + c} \in [0, 1]$, this becomes

$$g\left(\frac{c(n - 1) + (n + c)}{1 + c}\right) \leq \frac{c}{1 + c}g(n - 1) + \frac{1}{1 + c}g(n + c).$$

Note that the left-hand side is just $g(n)$. Therefore, this is the same as

$$c[g(n) - g(n - 1)] \leq g(n + c) - g(n).$$

This is exactly the inequality on the left to be proved.

For the other inequality, we simply put $y = n + 1$ and $z = n$ in (1). This gives

$$g(t(n + 1) + (1 - t)n) \leq tg(n + 1) + (1 - t)g(n).$$

Note that the left-hand side is just $g(n + t)$. This is exactly the other desired inequality.

- (b) Since $g(x) = \log f(x)$, the inequality in part (a) can be rewritten as

$$t(\log f(n) - \log f(n - 1)) \leq \log f(n + t) - \log f(n) \leq t(\log f(n + 1) - \log f(n)).$$

This implies

$$\left(\frac{f(n)}{f(n - 1)}\right)^t \leq \frac{f(n + t)}{f(n)} \leq \left(\frac{f(n + 1)}{f(n)}\right)^t.$$

As $f(n) = (n - 1)f(n - 1)$ and $f(n + 1) = nf(n)$, this can be reduced to

$$(n - 1)^t \leq \frac{f(n + t)}{f(n)} \leq n^t. \quad (2)$$

Now, note that $f(2) = 1 \cdot f(1) = 1$. Putting $n = 2$ and $t = \frac{1}{2}$ in (2), we obtain

$$1 \leq f\left(\frac{5}{2}\right) \leq \sqrt{2}.$$

Using $f\left(\frac{5}{2}\right) = \frac{3}{2}f\left(\frac{3}{2}\right) = \frac{3}{4}f\left(\frac{1}{2}\right)$, we get

$$\frac{4}{3} \leq f\left(\frac{1}{2}\right) \leq \frac{4}{3}\sqrt{2}.$$

Year 1999

1. r can be any number of the form $1 + \frac{1}{k}$, where k is an integer different from 0, -1.

If $r > 1$, let $r = 1 + \frac{m}{n}$ where $m, n \in \mathbb{Z}^+$ and $(m, n) = 1$. Since

$$r^{\frac{1}{r-1}} = \left(\frac{n+m}{n} \right)^{\frac{n}{m}}$$

and $(n+m, n) = (m, n) = 1$, we must have $n+m = a^m$ and $n = b^m$ for some $a, b \in \mathbb{Z}^+$. This gives $a^m - b^m = m$. Clearly, $a \geq b+1$. Therefore, we have

$$m = a^m - b^m = (a-b)(a^{m-1} + a^{m-2}b + \dots + b^{m-1}) \geq (1)(1+1+\dots+1) = m.$$

Equality must hold. This means $m = 1$ (since otherwise $a = b = 1$, contradiction) and $a - b = 1$. Conversely, when $m = 1$, $r = 1 + \frac{1}{n}$, so that $r^{\frac{1}{r-1}} = \left(\frac{n+1}{n} \right)^n \in \mathbb{Q}$.

If $r < 1$, let $r = 1 - \frac{m}{n}$ where $m, n \in \mathbb{Z}^+$ and $(m, n) = 1$. Since

$$r^{\frac{1}{r-1}} = \left(\frac{n-m}{n} \right)^{-\frac{n}{m}} = \left(\frac{n}{n-m} \right)^{\frac{n}{m}}$$

and $(n, n-m) = (n, -m) = 1$, we must have $n = a^m$ and $n-m = b^m$ for some $a, b \in \mathbb{Z}^+$. This gives $a^m - b^m = m$. Again, the only solution is $m = 1$ and $a - b = 1$.

When $m = 1$, $r = 1 - \frac{1}{n}$. Since $r > 0$, we need $n > 1$. In that case, we have $r^{\frac{1}{r-1}} = \left(\frac{n}{n-1} \right)^n \in \mathbb{Q}$.

Therefore, $r = 1 + \frac{1}{n}$ for $n \geq 1$ or $r = 1 - \frac{1}{n}$ for $n > 1$.

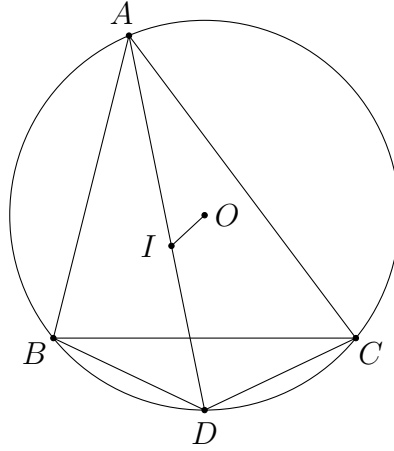
2. Let AI meet (ABC) again at D . Recall that $DB = DI = DC$. Applying Ptolemy's theorem to $ABCD$, we obtain

$$AD \times BC = AB \times CD + AC \times BD = DI(AB + AC).$$

Note that $\angle AIO \leq 90^\circ$ if and only if $AI \geq ID$. This is equivalent to

$$2 \leq \frac{AD}{DI} = \frac{AB + AC}{BC}.$$

This holds if and only if $2BC \leq AB + AC$ as desired.



3. The smallest n is 8.

We use terminologies in set theory. Let $A_1, A_2, \dots, A_n$ be sets corresponding to the n subjects, while the elements correspond to the students getting the best score in that subject. It is given that $|A_j| = 3$ and $|A_i \cap A_j| = 1$ for any $1 \leq i < j \leq n$. Suppose $n \geq 8$. We shall show that $|A_1 \cap A_2 \cap \dots \cap A_n| \geq 1$.

WLOG assume 1 is an element that belongs to the most number of sets. Suppose $1 \in A_j$ for $j = 1, 2, \dots, m$ and $1 \notin A_j$ for $j = m + 1, m + 2, \dots, n$. Suppose on the contrary that $m < n$.

Consider $A_n = \{a, b, c\}$. For $j = 1, 2, \dots, m$, since $|A_n \cap A_j| = 1$, each A_j consists of one of a, b, c . Also, each of a, b, c belongs to at most one of these A_j 's, since the only common element in these sets is 1, but $1 \notin A_n$. Thus, $m \leq 3$. This means each element belongs to at most 3 sets.

Now, suppose $A_1 = \{1, 2, 3\}$. Since each of 1, 2, 3 belongs to at most two other sets, we must have $n \leq 1 + 2 \times 3 = 7$, which is a contradiction. Therefore, $|A_1 \cap A_2 \cap \dots \cap A_n| \geq 1$.

It suffices to provide an example for $n = 7$ such that $A_1 \cap A_2 \cap \dots \cap A_n$ is empty, since we can remove some sets if n is less than 7. Indeed, consider

$$A_1 = \{1, 2, 3\}, A_2 = \{1, 4, 5\}, A_3 = \{1, 6, 7\}, A_4 = \{2, 4, 6\},$$

$$A_5 = \{2, 5, 7\}, A_6 = \{3, 4, 7\}, A_7 = \{3, 5, 6\}.$$

It is routine to check that $|A_i \cap A_j| = 1$ for all i, j , but no element belongs to all of them.

4. The solutions are $f(x) = 0$ for all $x \in \mathbb{R}$ and $f(x) = x$ for all $x \in \mathbb{R}$.

It is easy to check that both are solutions. Label the equation as follows.

$$f(x + yf(x)) = f(x) + xf(y) \quad (1)$$

Putting $x = 1$ in (1), we have

$$f(1 + yf(1)) = f(1) + f(y). \quad (2)$$

If $f(1) \neq 1$, then there exists $y \in \mathbb{R}$ such that $1 + yf(1) = y$. For this y , we obtain $f(1) = 0$. Then (2) becomes $f(y) = 0$ for all y , which is the first solution.

Now, we assume $f(1) = 1$. If $f(t) = 0$ for some $t \in \mathbb{R}$, by putting $x = t$ and $y = 1$ in (1), we obtain $t = 0$. This means $f(x) \neq 0$ for any $x \neq 0$. Putting $x = 1$ in (1), we have

$$f(y + 1) = f(y) + 1. \quad (3)$$

By induction we easily obtain $f(n) = n$ for any $n \in \mathbb{Z}$. Putting $x = -1$ in (1), we find that

$$f(-y - 1) = -f(y) - 1.$$

Using (3), we get $-f(y) = f(-y - 1) + 1 = f(-y)$. Replacing y by $-y$ in (1), we have

$$f(x - yf(x)) = f(x) - xf(y).$$

Adding this to (1), we have

$$f(x + yf(x)) + f(x - yf(x)) = 2f(x).$$

For $x \neq 0$, we can replace y by $\frac{y}{f(x)}$. This yields

$$f(x + y) + f(x - y) = 2f(x). \quad (4)$$

Note that this also holds when $x = 0$. Considering $x = y$, we find that

$$f(2x) = 2f(x).$$

Also, applying the substitution $a = x + y$ and $b = x - y$, equation (4) becomes

$$f(a) + f(b) = 2f\left(\frac{a+b}{2}\right) = f(a+b).$$

This shows f satisfies the Cauchy equation. Thus, (1) can be simplified to

$$f(yf(x)) = xf(y). \quad (5)$$

Putting $y = 1$, we obtain $f(f(x)) = x$. Replacing x by $f(x)$ in (5) and using $f(f(x)) = x$, we get

$$f(xy) = f(x)f(y). \quad (6)$$

It is well-known that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the Cauchy equation and (6) can only be the zero function or the identity function. Thus, another solution is $f(x) = x$ for all $x \in \mathbb{R}$.

Year 2000

1. Note that OE is the perpendicular bisector of AB . Let $\angle EBA = \angle EAB = x$. Similarly, let $\angle FCA = \angle FAC = y$. Then we have

$$\angle BPC = 180^\circ - \angle PBC - \angle PCB = 2x + 2y = 2\angle BAC = \angle BOC.$$

This implies O, B, C, P are concyclic. Applying Ptolemy's theorem to the cyclic quadrilateral $OBCP$, we obtain

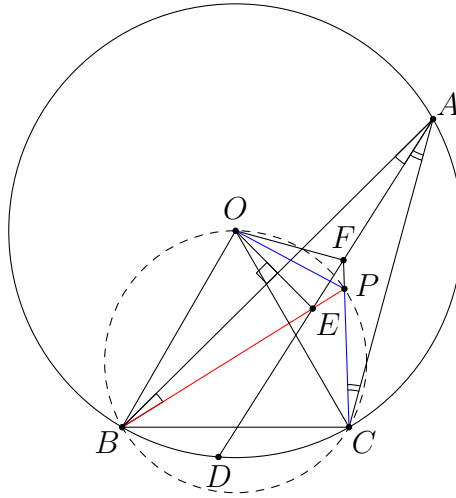
$$OB \times PC + BC \times OP = OC \times BP.$$

Thus,

$$BC \times OP = OC \times BP - OB \times PC = OB(BP - PC) = OB \times OP.$$

This yields $BC = OB$. Hence, $\triangle OBC$ is equilateral. Therefore,

$$\angle BAC = \frac{1}{2} \angle BOC = 30^\circ.$$



2. The answer is 44.

We shall prove by induction that $\sqrt{n} < a_n < b_n := \frac{n}{\sqrt{n-1}}$ for any integer $n \geq 3$.

For the base case, we find $a_2 = 2$ and $a_3 = 2 \in \left[\sqrt{3}, \frac{3}{\sqrt{2}} \right]$.

For the inductive step, assume $\sqrt{k} < a_k < b_k$ for some integer $k \geq 3$. Note that $b_k < k$. Therefore, $\frac{a_k}{k} < \frac{b_k}{k} < 1$. Observe that the function $f(x) = x + \frac{1}{x}$ is

decreasing on $(0, 1)$ since $f'(x) = 1 - \frac{1}{x^2} < 0$. Therefore,

$$a_{k+1} = f\left(\frac{a_k}{k}\right) < f\left(\frac{\sqrt{k}}{k}\right) = \frac{k+1}{\sqrt{k}} = b_{k+1}.$$

For the lower bound, we have

$$a_{k+1} = f\left(\frac{a_k}{k}\right) > f\left(\frac{b_k}{k}\right) = \frac{k}{\sqrt{k-1}} > \sqrt{k+1}.$$

So we have proven $\sqrt{n} < a_n < b_n$ for all $n \geq 3$ by induction. This implies

$$a_n < \frac{n}{\sqrt{n-1}} = \sqrt{n+1 + \frac{1}{n-1}} < \sqrt{n+2}.$$

Thus, we have

$$44 < \sqrt{2000} < a_{2000} < \sqrt{2002} < 45.$$

This shows the greatest integer less than or equal to a_{2000} is 44.

3. The solutions are $(p, q) = (5, 5), (5, 11), (5, 61)$ and their permutations.

If $p \mid 7^p - 2^p$, then $p \mid 7 - 2 = 5$ by the Fermat little theorem. This implies $p = 5$. If we further have $q \mid 7^q - 2^q$, then $q = 5$ for the same reason. So $p = q = 5$ is a solution. If $q \neq 5$, then $q \mid 7^p - 2^p = 16775 = 5^2 \times 11 \times 61$. Therefore, $q = 11, 61$. By symmetry $(p, q) = (11, 5), (61, 5)$ are also solutions.

Now we may assume $p, q \neq 5$. Then we must have $p \mid 7^q - 2^q$ and $q \mid 7^p - 2^p$. WLOG assume $q \geq p$. Consider $p \mid 7^q - 2^q$. Note that $(q, p-1) = 1$ since q is a prime and $q > p-1$. Thus, there exists $a \in \mathbb{Z}^+$ such that $aq \equiv 1 \pmod{p-1}$. By the Fermat little theorem, we have

$$7 \equiv 7^{aq} = (7^q)^a \equiv (2^q)^a = 2^{aq} \equiv 2 \pmod{p}.$$

This forces $p = 5$, which is a contradiction.

Therefore, the only solutions are $(p, q) = (5, 5), (5, 11), (5, 61)$ and their permutations.

4. n can be any even integer larger than 2.

We first consider the case when n is odd. Let (x_j, y_j) , where $j = 1, 2, \dots, n$, be the vertices of the polygon in anticlockwise direction. WLOG assume $x_1 = y_1 = 0$. By using the homothety $(x_j, y_j) \mapsto \left(\frac{x_j}{d}, \frac{y_j}{d}\right)$, we may assume the greatest common divisor of $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ is 1. In particular, not all coordinates are even numbers.

Let $a_j = x_{j+1} - x_j$ and $b_j = y_{j+1} - y_j$ for $j = 1, 2, \dots, n$, where the indices are taken modulo n . It is given that $a_j^2 + b_j^2 = c$ for some constant c .

- If $c \equiv 0 \pmod{4}$, then both a_j, b_j are even. Since x_0, y_0 are even, we easily deduce that all x_j, y_j are even. This contradicts our assumption.
- If c is odd, then $a_j + b_j \equiv 1 \pmod{2}$. Thus, we have

$$1 \equiv \sum_{j=1}^n (a_j + b_j) = \sum_{j=1}^n (x_{j+1} - x_j + y_{j+1} - y_j) = 0 \pmod{2},$$

which is a contradiction.

- If $c \equiv 2 \pmod{4}$, then a_j, b_j are odd. Thus, we have

$$1 \equiv \sum_{j=1}^n a_j = \sum_{j=1}^n (x_{j+1} - x_j) = 0 \pmod{2},$$

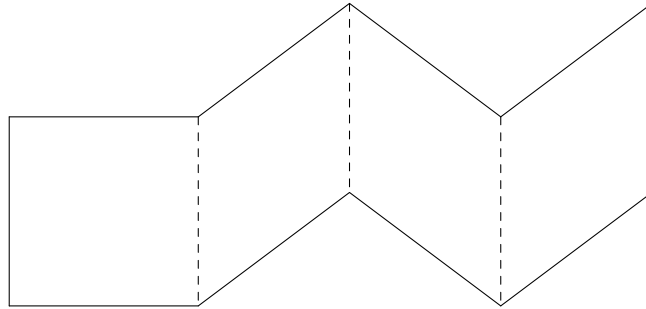
which is a contradiction.

Therefore, it is impossible that n is odd.

For even $n \geq 4$, we provide the following construction. The following quadrilaterals have the properties that each side has length 5, and two of the sides are vertical.

- (type I) a square with vertices $(0, 0), (5, 0), (5, 5), (0, 5)$
- (type II) a rhombus with vertices $(0, 0), (4, 3), (4, 8), (0, 5)$
- (type III) a rhombus with vertices $(0, 0), (4, -3), (4, 2), (0, 5)$

We can place $\frac{n}{2} - 1$ copies of these quadrilaterals in the order types I, II, III, II, III, II, III, $\dots$ by translating them to the position such that the left edge of the next quadrilateral overlaps with the right edge of the previous one as shown. This gives an n -sided polygon whose every side has length 5.



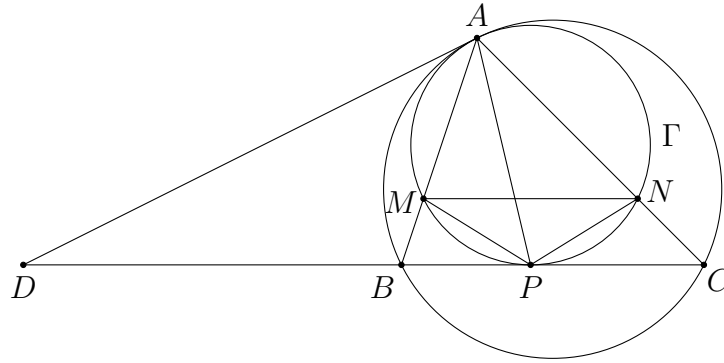
Year 2001

1. If $AB = AC$, the result is obvious due to symmetry (both statements are equivalent to P being the midpoint of BC). WLOG assume $AB < AC$. Let D be the intersection of the line BC and the tangent at A to (ABC) .

If $\widehat{MP}$ and $\widehat{NP}$ are equal, then MN is parallel to the tangent at P to Γ , which is BC . It follows that

$$\angle DAM = \angle DAB = \angle ACB = \angle ANM,$$

which implies DA is tangent to Γ . Therefore, Γ is tangent to (ABC) at A .



Conversely, if Γ and (ABC) are tangent at A , then DA is tangent to Γ . Therefore, we have

$$\angle ANM = \angle DAM = \angle DAB = \angle ACB,$$

which implies $MN \parallel BC$. This shows P is the midpoint of $\widehat{MN}$ as desired.

2. n can be 1 or 3.

WLOG assume $x \geq y \geq z$. Then we have

$$3x^3 \geq x^3 + y^3 + z^3 = nx^2y^2z^2,$$

which gives $x \geq \frac{n}{3}y^2z^2$.

Next, we rewrite the equation as $y^3 + z^3 = x^2(ny^2z^2 - x)$. Since the left-hand side is positive, we have $ny^2z^2 - x \geq 1$. Thus,

$$2y^3 \geq y^3 + z^3 \geq \left(\frac{n}{3}y^2z^2\right)^2 (1) = \frac{n^2}{9}y^4z^4.$$

This yields $18 \geq n^2yz^4$. There is no positive integer solution when $n \geq 5$. When $n = 4$, we must have $y = z = 1$. The given equation becomes $x^3 + 2 = 4x^2$. This shows $x^2 \mid 2$, and hence $x = 1$. But this is not a solution.

For $n = 3$, we can take $x = y = z = 1$.

For $n = 2$, we have $18 \geq 4yz^4$. So $z = 1$ and $y \leq 4$. Note that $x^2 \mid y^3 + z^3 = y^3 + 1$. As $x \geq y$, (x, y) can only be $(1, 1), (3, 2)$. It is routine to check that none of these is a solution to $x^3 + y^3 + 1 = 2x^2y^2$.

For $n = 1$, we can take $(x, y, z) = (3, 2, 1)$.

Thus, n can only be 1 or 3.

3. Note that $(k+1) - 0 \mid F(k+1) - F(0)$. Since $|F(k+1) - F(0)| \leq k$, we must have $F(k+1) = F(0) = d$ for some constant d . Let $F(x) = d + x(x-k-1)G(x)$ for some polynomial G with integer coefficients.

For $2 \leq n \leq k-1$, we have $F(n) = d + n(n-k-1)G(n)$, and so $n(k+1-n) \mid F(n) - d$. Again, $|F(n) - d| \leq k$. Since $n(k+1-n) \geq 2(k+1-2) > k$, we must have $F(n) = d$. Thus, we can write $F(x) = d + x(x-2)(x-k+1)(x-k-1)H(x)$ for some polynomial H with integer coefficients. (Note that we have used the fact $k \geq 4$ to show that the roots $0, 2, k-1, k+1$ of $F(x) - d$ are distinct.)

Lastly, for $n = 1, k$, we have $n(n-2)(n-k+1)(k+1-n) \mid F(n) - d$. Again, $|F(n) - d| \leq k$. Also, $n(n-2)(n-k+1)(k+1-n) = k(k-2) \geq 2k$ in both cases. Therefore, we must have $F(n) = d$.

Thus, we have shown that $F(n) = d$ for $n = 0, 1, \dots, k+1$.

4. Let A, B, C, D, E, F be 6 distinct points on the circle such that

$$\widehat{AB} = \widehat{BC} = \widehat{CD} = \widehat{DE} = \widehat{EF} = \widehat{FA}.$$

Let O be the centre of the circle. Note that any two points in the same sector among $AOB, BOC, COD, DOE, EOF, FOA$ have distance at most 1. Let $n_1, n_2, \dots, n_6$ be the number of points in the 6 sectors respectively. It suffices to show

$$\binom{n_1}{2} + \binom{n_2}{2} + \dots + \binom{n_6}{2} \geq 2001.$$

Indeed, since the binomial function $\binom{x}{2}$ is convex, by Jensen's inequality, we have

$$\sum_{k=1}^6 \binom{n_k}{2} \geq 6 \binom{\frac{n_1+n_2+\dots+n_6}{6}}{2} > 6 \binom{35}{2} = 3570 > 2001.$$

This completes the proof.

Year 2002

1. (a) We only provide a proof for the configuration as shown. The other cases are similar.

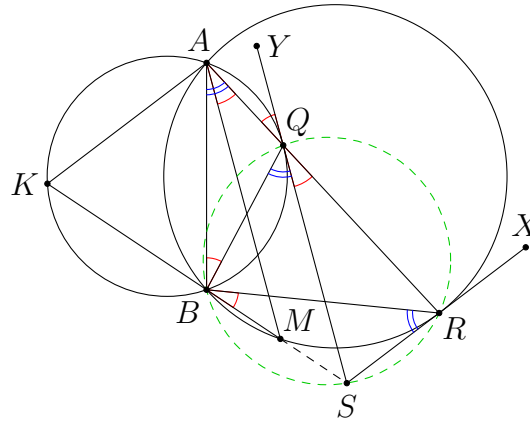
Let X and Y be any points on the tangent at R to (ABR) and the tangent at Q to (ABQ) respectively as shown. We have

$$\angle MBR = \angle MAR = \angle YQA = \angle QBA,$$

and so

$$\begin{aligned}\angle XRA &= \angle RBA = \angle RBQ + \angle QBA = \angle RBQ + \angle MBR \\ &= \angle MBQ = \angle KAQ = \angle KAR.\end{aligned}$$

This implies $RX \parallel KA$.



- (b) Let the tangent at R to (ABR) and the tangent at Q to (ABQ) meet at S . Since

$$\angle BRS = \angle BAR = \angle BAQ = \angle BQS,$$

the points Q, B, S, R are concyclic. This implies

$$\angle SBR = \angle SQR = \angle MBR,$$

and hence S, M, B are collinear. Thus, the two tangents and KM are concurrent.

2. n can be any odd integer larger than 1.

We use terminologies in graph theory. We need to colour all edges of the complete graph of n vertices in n colours $C_1, C_2, \dots, C_n$ such that for any three distinct colours, there exists a triangle whose edges are of these three colours.

Note that there are $\binom{n}{3}$ triangles as well as $\binom{n}{3}$ triples of colours. Therefore, the triangles and the triples of colours are in one-to-one correspondence.

For each colour C_j , there are $\binom{n-1}{2}$ triples of colours consisting of C_j since we need to choose two more colours from the remaining $n-1$ colours. On the other hand, for every edge of colour C_j , there are $n-2$ triangles containing this edge since we need to choose one more vertex from the remaining $n-2$ vertices. Therefore, there are exactly

$$\frac{1}{n-2} \binom{n-1}{2} = \frac{n-1}{2}$$

edges of colour C_j . This implies n must be odd.

We now provide a colouring for odd n . For any vertices V_i and V_j , we use colour C_r to colour the edge joining them where $r \equiv i + j \pmod{n}$. It suffices to show that for any distinct triples r_1, r_2, r_3 , the system

$$\begin{cases} i + j \equiv r_1 \pmod{n}, \\ j + k \equiv r_2 \pmod{n}, \\ k + i \equiv r_3 \pmod{n} \end{cases}$$

has a unique solution in $i, j, k \pmod{n}$ such that i, j, k are pairwise incongruent modulo n . Indeed, we readily find that

$$i \equiv 2^{-1}((i+j) - (j+k) + (k+i)) \equiv 2^{-1}(r_1 - r_2 + r_3) \pmod{n},$$

and similarly $j \equiv 2^{-1}(r_1 + r_2 - r_3) \pmod{n}$ and $k \equiv 2^{-1}(-r_1 + r_2 + r_3) \pmod{n}$. These are incongruent because r_1, r_2, r_3 are incongruent modulo n . This completes the proof.

3. Let $f(a, b, c) = ab^2 + bc^2 + ca^2$. Note that it suffices to show

$$f(a, b, c) - f(a, c, b) \leq 0, \tag{1}$$

$$f(a, b, c) + f(a, c, b) \leq \frac{27}{4}, \tag{2}$$

since adding these yields the result. To prove (1), observe that

$$f(a, b, c) - f(a, c, b) = (a-b)(b-c)(c-a) \leq 0.$$

To prove (2), by homogenizing the inequality, we need to prove

$$ab^2 + bc^2 + ca^2 + a^2b + b^2c + c^2a \leq \frac{1}{4}(a+b+c)^3.$$

This is equivalent to

$$a^3 + b^3 + c^3 + 6abc \geq ab^2 + bc^2 + ca^2 + a^2b + b^2c + c^2a,$$

which holds by Schur's inequality and $3abc \geq 0$. Equality holds when $c = 0$ and $a = b = \frac{3}{2}$.

4. The answer is $\frac{p-1}{4}$.

Note that $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ are pairwise incongruent modulo p since

$$x^2 \equiv y^2 \pmod{p} \Rightarrow x \equiv \pm y \pmod{p}.$$

As there are exactly $\frac{p-1}{2}$ nonzero quadratic residues, $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ are all the nonzero quadratic residues modulo p .

Since $p \equiv 1 \pmod{4}$, -1 is a quadratic residue modulo p . Therefore, r is a quadratic residue modulo p if and only if $-r$ is a quadratic residue modulo p . This shows the nonzero quadratic residues come in pairs such that the numbers in each pair are negative of each other modulo p . Thus, $\left\{\frac{k^2}{p}\right\}$ for $k = 1, 2, \dots, \frac{p-1}{2}$ can be partitioned into $\frac{p-1}{4}$ pairs, such that the sum of each pair is $\frac{r + (p-r)}{p} = 1$. Thus, the sum is equal to $\frac{p-1}{4}$.

Year 2003

1. The greatest K is 16.

We first prove the inequality when $K = 16$. Note that

$$\begin{aligned} u^2 - 4vw &= u^2 + 2vw - 6vw \\ &\geq u^2 + 2vw - 3(v^2 + w^2) \\ &= u^2 + (v + w)^2 - 4(v^2 + w^2) \\ &\geq 2u(v + w) - 4(v^2 + w^2) \\ &= 2((uw - 2v^2) + (uv - 2w^2)). \end{aligned}$$

If $uw - 2v^2$ and $uv - 2w^2$ have different signs (possibly equal to 0), we have

$$(u^2 - 4vw)^2 \geq 0 \geq K(2v^2 - uw)(2w^2 - uv).$$

The equality case will be handled below.

If both $uw - 2v^2$ and $uv - 2w^2$ are negative, then $(uw)(uv) < (2v^2)(2w^2)$, which yields the contradiction $u^2 < 4vw$.

Therefore, we may assume both $uw - 2v^2$ and $uv - 2w^2$ are positive. Therefore, we obtain

$$(u^2 - 4vw)^2 \geq 4((uw - 2v^2) + (uv - 2w^2))^2 \geq 16(uw - 2v^2)(uv - 2w^2).$$

For the equality to hold, we must have $v = w$ and $u = v + w$. This implies

$$u^2 = (2v)^2 = 4v^2 = 4vw,$$

contradicting $u^2 > 4vw$. Therefore, equality cannot hold, and the inequality is proven.

Next, we show $K = 16$ is the largest possible. Consider $v = w = 1$ and $u = 2 + \varepsilon$ for a sufficiently small positive ε . The relation $u^2 > 4vw$ holds. Also, we have

$$(u^2 - 4vw)^2 = (4 + 4\varepsilon + \varepsilon^2 - 4)^2 = \varepsilon^2(4 + \varepsilon)^2$$

and

$$K(2v^2 - uw)(2w^2 - uv) = K(2 - 2 - \varepsilon)^2 = K\varepsilon^2.$$

The inequality holds if and only if $(4 + \varepsilon)^2 > K$. When ε approaches 0, the left-hand side approaches 16. Therefore, $K \leq 16$. This completes the proof.

2. The answer is $\frac{7 - \sqrt{7}}{14}(1 + \sqrt{7})^{2003} + \frac{7 + \sqrt{7}}{14}(1 - \sqrt{7})^{2003}$.

For each integer $n \geq 0$, let a_n be the number of paths of length n starting at O and terminating at O . Also, let b_n be the number of paths of length n starting

To count a_n for $n \geq 1$, the $(n-1)$ st step must terminate at one of A, B, C, D, E, F . For each of these 6 points, there are b_{n-1} paths terminating at that point. This gives a total of $6b_{n-1}$ paths. In other words, we have

To count b_n for $n \geq 1$, suppose the n th step terminates at A . Then the $(n-1)$ st step may terminate at B, F or O . If it is B or F , then there are b_{n-1} paths. If it is O , then there are a_{n-1} paths. Therefore, we have

Combining (1) and (2), we obtain

This can be rewritten as $a_n = 2a_{n-1} + 6a_{n-2}$ for $n \geq 2$. The roots of the characteristic equation $\lambda^2 - 2\lambda - 6 = 0$ are $1 \pm \sqrt{7}$. Suppose $a_n = A(1 + \sqrt{7})^n + B(1 - \sqrt{7})^n$. Using the initial conditions $a_0 = 1$ and $a_1 = 0$, we solve

This yields $A = \frac{7 - \sqrt{7}}{14}$ and $B = \frac{7 + \sqrt{7}}{14}$. Therefore, the answer is

3.



Let H_1, H_2, H_3, H_4 be the orthocentres of $\triangle AKN$, $\triangle BKL$, $\triangle CLM$, $\triangle DMN$ respectively, and let O be the centre of $(ABCD)$.

Since K is the midpoint of AB , we have $OK \perp AB$. Thus, $OK \parallel H_1N$. Similarly, $ON \parallel H_1K$. This shows H_1KON is a parallelogram. By symmetry, H_4NOM is a parallelogram. This shows $H_1K = NO = H_4M$ and $H_1K \parallel NO \parallel H_4M$. Therefore, $H_1KM H_4$ is a parallelogram. By symmetry, KH_2H_3M is a parallelogram. This shows $H_1H_4 = KM = H_2H_3$ and $H_1H_4 \parallel KM \parallel H_2H_3$. Therefore, $H_1H_2H_3H_4$ is a parallelogram.

4. The solutions are $(a, b, c) = (1, 0, 0), (2, -1, -1)$ up to permutation.

Let $s = a + b + c$. Consider

$$P(x) = (x - a)(x - b)(x - c) = x^3 - sx^2 + (ab + bc + ca)x - abc.$$

Putting $x = s$, we obtain $(b + c)(c + a)(a + b) = (ab + bc + ca)s - abc$. Therefore, the given equation becomes

$$(ab + bc + ca)s - abc + 2s^3 = 2 - 2abc.$$

This is the same as $-s^3 - s^3 - (ab + bc + ca)s - abc = -2$, i.e. $P(-s) = -2$. In other words, we have

$$(2a + b + c)(a + 2b + c)(a + b + 2c) = 2.$$

(One can prove this by direct expansion.) By symmetry, it suffices to consider a few cases as follows.

- If $2a + b + c = 2$ and $a + 2b + c = a + b + 2c = 1$, adding these we obtain $4(a + b + c) = 4$. So we have $a + b + c = 1$, and hence $(a, b, c) = (1, 0, 0)$.
- If $2a + b + c = 2$ and $a + 2b + c = a + b + 2c = -1$, adding these we obtain $4(a + b + c) = 0$. So we have $a + b + c = 0$, and hence $(a, b, c) = (2, -1, -1)$.
- If $2a + b + c = -2$, $a + 2b + c = \pm 1$ and $a + b + 2c = \mp 1$, adding these we obtain $4(a + b + c) = -2$. There is no integer solution.

This shows the only solutions are $(a, b, c) = (1, 0, 0), (2, -1, -1)$ up to permutation.

Year 2004

1. Let $d = a_j - a_{j-1} \geq 0$. If $d > 0$, then

$$\frac{1}{a_k^2} < \frac{1}{a_k^2 - d^2} = \frac{1}{a_{k-1}a_{k+1}} = \frac{a_{k+1} - a_{k-1}}{2da_{k-1}a_{k+1}} = \frac{1}{2d} \left(\frac{1}{a_{k-1}} - \frac{1}{a_{k+1}} \right)$$

for any $k > 1$. Therefore, we have

$$\begin{aligned} \sum_{k=2}^n \frac{1}{a_k^2} &< \frac{1}{2d} \sum_{k=2}^n \left(\frac{1}{a_{k-1}} - \frac{1}{a_{k+1}} \right) \\ &= \frac{1}{2d} \left(\frac{1}{a_1} - \frac{1}{a_n} + \frac{1}{a_2} - \frac{1}{a_{n+1}} \right) \\ &= \frac{1}{2d} \left(\frac{(n-1)d}{a_1a_n} + \frac{(n-1)d}{a_2a_{n+1}} \right) \\ &= \frac{n-1}{2} \cdot \frac{a_1a_n + a_2a_{n+1}}{a_1a_2a_na_{n+1}}. \end{aligned}$$

When $d = 0$, let $a_j = c$ for all j . Then

$$\sum_{k=2}^n \frac{1}{a_k^2} = \frac{n-1}{c^2}$$

and

$$\frac{n-1}{2} \cdot \frac{a_1a_n + a_2a_{n+1}}{a_1a_2a_na_{n+1}} = \frac{n-1}{2} \cdot \frac{2c^2}{c^4} = \frac{n-1}{c^2} = \sum_{k=2}^n \frac{1}{a_k^2}.$$

Therefore, the inequality is proven, and equality holds when $a_1 = a_2 = \dots = a_{n+1}$.

2. We count the number of triples (T, S_1, S_2) such that T is a teacher teaching two distinct students S_1 and S_2 . Note that (T, S_1, S_2) is different from (T, S_2, S_1) .

For each of the b teachers, there are k choices for S_1 and $k-1$ choices for S_2 . Therefore, there are $bk(k-1)$ such triples.

For each of the c students S_1 and each of the $c-1$ students S_2 , there are h choices for T . Therefore, there are $c(c-1)h$ such triples.

Thus, we have obtained $bk(k-1) = c(c-1)h$. This clearly implies $\frac{b}{h} = \frac{c(c-1)}{k(k-1)}$.

3. Let PS and QR meet BC at X and Y respectively. Note that A, P, X, D are concyclic since $\angle ADX + \angle APX = 180^\circ$. Similarly, A, D, Y, Q are concyclic. Also, since

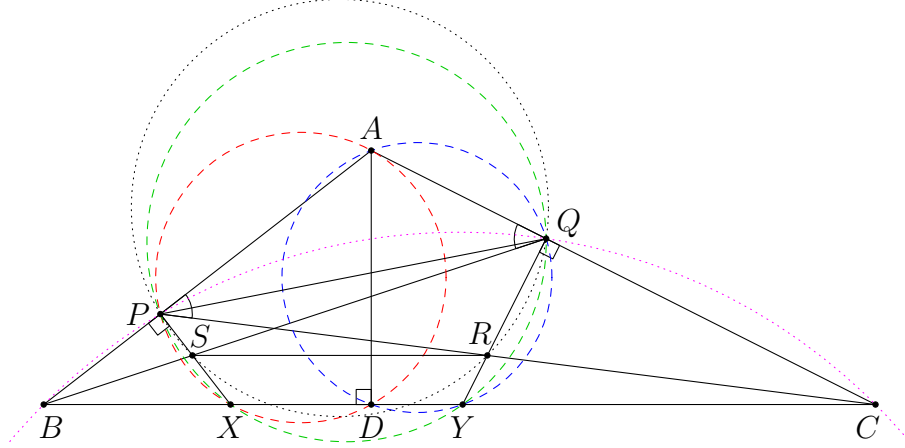
$$\angle BPC = 180^\circ - \angle CPA = 180^\circ - \angle AQB = \angle BQC,$$

the points B, C, Q, P are concyclic. Thus, we have

$$\angle PQY = 90^\circ - \angle AQP = 90^\circ - \angle CBP = \angle PXB.$$

This implies Q, P, X, Y are concyclic.

Consider the circles $(APXD)$, $(ADYQ)$ and $(QPXY)$. Their pairwise radical axes are AD, PX, QY respectively. Clearly, these lines are not parallel. Therefore, they are concurrent at the radical centre of these circles.



Next, observe that Q, P, S, R are concyclic since

$$\angle SQR = 90^\circ - \angle AQS = 90^\circ - \angle RPA = \angle SPR.$$

Applying Reim's theorem to $(QPXY)$ and $(QPSR)$, we obtain $SR \parallel XY$. (It simply follows from $\angle XSR = \angle PQR = 180^\circ - \angle YXS$.)

4. There are 348364800 such functions.

We first claim that condition (iii) can be replaced by the completely multiplicative condition. This means

$$f(p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s}) = f(p_1)^{a_1} f(p_2)^{a_2} \cdots f(p_s)^{a_s}$$

for any distinct primes $p_1, p_2, \dots, p_s$ and positive integers $a_1, a_2, \dots, a_s$. We prove this by induction on $a_1 + a_2 + \cdots + a_s$.

The base cases $a_1 + a_2 + \cdots + a_s = 0, 1$ hold trivially. Assume the result holds whenever the sum of exponents is less than k . Consider $n = p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s}$ where $a_1 + a_2 + \cdots + a_s = k$. Note that n is composite, and so $g(n), h(n) > 1$. Since $g(n)h(n) = n$, the sum of exponents in the prime factorizations of $g(n)$ and $h(n)$ are both smaller than k . Thus, by the inductive hypothesis, $f(g(n))$ and $f(h(n))$ can be decomposed into a product of $f(p_j)$'s. Clearly, this proves

$$f(n) = f(p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s}) = f(p_1)^{a_1} f(p_2)^{a_2} \cdots f(p_s)^{a_s}.$$

By induction, we have shown that f is completely multiplicative. Conversely, if f is completely multiplicative, condition (iii) holds trivially.

Let $\mathcal{P}$ be the set of all primes from 1 to 100, and let $\mathcal{C} = S - (\mathcal{P} \cup \{1\})$ be the set of all composites. For any $n \in \mathcal{C}$, since $g(n), h(n) > 1$, we must have $f(g(n)), f(h(n)) > 1$. Therefore, $f(n) = f(g(n))f(h(n)) \in \mathcal{C}$. This implies $f(\mathcal{C}) = \mathcal{C}$. As $f(1) = 1$, by the bijective condition, we must have $f(\mathcal{P}) = \mathcal{P}$. Thus, it remains to assign distinct prime values to $f(p)$ for $p \in \mathcal{P}$. Afterwards we can easily compute $f(n)$ by the completely multiplicative condition, while the bijective condition is always satisfied.

Now, the only constraint is that $f(n) \in S$ for any $n \in S$. For example, since

$$100 \geq f(64) = f(2)^6$$

and $f(2) \neq 1$, we must have $f(2) = 2$. Similarly, by considering $f(81)$, we can prove that $f(3) = 3$. Next, since

$$100 \geq f(75) = f(3)f(5)^2,$$

we have $f(5) \leq 5$, and hence $f(5) = 5$. Similarly, by considering $f(98), f(99)$ and $f(91)$, we can prove that $f(p) = p$ for $p = 7, 11, 13$.

Next, by considering $f(85)$ and $f(95)$, we can only prove that $f(17), f(19) \leq 20$. Therefore, we have $\{f(17), f(19)\} = \{17, 19\}$. There is no further constraint on $f(17)$ and $f(19)$ since we can verify that $f(n) \in S$ for any n being multiples of 17 or 19, as long as $f(17), f(19) \leq 20$.

Similarly, by considering $f(92)$, we find that $f(23) = 23$.

By considering $f(87)$ and $f(93)$, we find that $\{f(29), f(31)\} = \{29, 31\}$.

By considering $f(74), f(82), f(86)$ and $f(94)$, we find that 37, 41, 43, 47 can be mapped to any permutation of themselves.

Lastly, the images of 53, 59, 61, 67, 71, 73, 79, 83, 89, 97 can be any permutation of themselves.

In view of the above arguments, we can choose the images within each group in an arbitrary way. Each choice corresponds to a unique function satisfying all requirements. Therefore, the number of such functions is

$$2! \times 2! \times 4! \times 10! = 348364800.$$

Year 2005

1. We use terminologies in graph theory. We shall prove by induction that in a complete graph of $3 \cdot k!$ vertices, if we colour every edge in one of k colours, then there exists a monochromatic triangle.

For the base case $k = 1$, since there are 3 vertices but only one colour, the result holds trivially.

Assume the claim holds for some k . Consider a complete graph of $3 \cdot (k + 1)!$ vertices. We colour every edge in one of $k + 1$ colours. Fix any vertex A . By the pigeonhole principle, there are

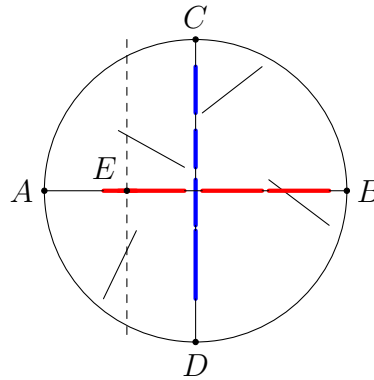
$$\left\lceil \frac{3 \cdot (k + 1)! - 1}{k + 1} \right\rceil = 3 \cdot k!$$

edges with A as an endpoint which are of the same colour, say red. If an edge connecting two of these vertices is red, then these two vertices together with A form a red triangle. If all these edges are not red, then they are in one of k colours. By the inductive hypothesis, there is a monochromatic triangle in this subgraph. This proves the inductive step, and hence the problem is solved.

2. Let AB and CD be the diameters of the circle which are parallel and perpendicular to L respectively. Let P_iQ_i be the projection of each segment on AB , and let X_iY_i be the projection of each segment on CD . Note that $P_iQ_i + X_iY_i \geq 1$. Therefore, we have

$$\sum_{j=1}^{4n} P_iQ_i + \sum_{j=1}^{4n} X_iY_i = \sum_{j=1}^{4n} (P_iQ_i + X_iY_i) \geq 4n = AB + CD.$$

WLOG assume $\sum_{j=1}^{4n} P_iQ_i \geq AB$. As all the segments P_iQ_i lie strictly inside the segment AB , two of these segments P_iQ_i and P_jQ_j must overlap. Let E be a point lying on both of them. Then the line passing through E and perpendicular to AB intersects two unit segments whose projections are P_iQ_i and P_jQ_j . This completes the proof.



3. We claim that $f(x) = 6x^3 - x^2 \geq \frac{5x-1}{8}$ for any $x > 0$. Indeed,

$$\begin{aligned} & 6x^3 - x^2 \geq \frac{5x-1}{8} \\ \Leftrightarrow & 48x^3 - 8x^2 - 5x + 1 \geq 0 \\ \Leftrightarrow & (4x-1)^2(3x+1) \geq 0. \end{aligned}$$

This clearly holds. It follows that

$$f(a) + f(b) + f(c) + f(d) \geq \frac{5(a+b+c+d) - 4}{8} = \frac{1}{8}.$$

Equality holds when $a = b = c = d = \frac{1}{4}$.

4. Firstly, note that $2005 \equiv 1 \pmod{3}$. Therefore, $3 \mid 2005 - 1$. Suppose we have chosen distinct primes $p_1, p_2, \dots, p_k$ such that $p_1 p_2 \cdots p_k \mid 2005^{p_1 p_2 \cdots p_{k-1}} - 1$, where the exponent of 2005 is 1 when $k = 1$. Then

$$2005^{p_1 p_2 \cdots p_k} = (2005^{p_1 p_2 \cdots p_{k-1}})^{p_k} \equiv 1^{p_k} = 1 \pmod{p_1 p_2 \cdots p_k}.$$

This shows $n = p_1 p_2 \cdots p_k$ satisfies the requirement.

By Zsigmondy's theorem, there exists a prime p_k dividing $2005^{p_1 p_2 \cdots p_k} - 1$ but not $2005^{p_1 p_2 \cdots p_{k-1}} - 1$. This shows p_k is different from $p_1, p_2, \dots, p_{k-1}$. Therefore, we have

$$p_1 p_2 \cdots p_{k+1} \mid 2005^{p_1 p_2 \cdots p_k} - 1.$$

By induction, we can find an infinite sequence of distinct primes p_j such that $n = p_1 p_2 \cdots p_k$ satisfies the requirement for any $k \in \mathbb{Z}^+$. So we are done.

Year 2006

1. The answer is 1004.

When $M = \{1003, 1004, \dots, 2006\}$, the sum of any two elements in M is at least 2007, which is larger than the largest element in M . Therefore, $x + y \nmid z$ for any $x, y, z \in M$. This gives a possible case for $|M| = 1004$.

Next, consider any subset M satisfying the constraint. Suppose the largest element of M is a . Then M contains at most one element in each of the pairs

$$(1, a-1), (2, a-2), \dots, \left(\left\lfloor \frac{a}{2} \right\rfloor, \left\lceil \frac{a}{2} \right\rceil\right)$$

(note that the last pair may contain the same number, which does not affect the validity of our claim). Therefore, M contains at most $\frac{a}{2}$ elements smaller than a . Since a is the largest element, M cannot contain elements larger than a . Therefore,

$$|M| \leq \frac{a}{2} + 1 \leq 1003 + 1 = 1004.$$

Hence, the largest possible size of M is 1004.

2. (IMO Shortlist 1990 modified) The answer is 169.

Firstly, since $2^{2006} < 10^{2006}$, we have

$$f(2^{2006}) \leq (9 \times 2006)^2 < 10^{10}.$$

Secondly, this implies

$$f_2(2^{2006}) \leq (9 \times 10)^2 = 8100.$$

Similarly, we find that

$$f_3(2^{2006}) \leq (7 + 9 + 9 + 9)^2 = 1156$$

and

$$f_4(2^{2006}) \leq (9 \times 3)^2 = 729.$$

It follows that the sum of digits of $f_k(2^{2006})$ cannot exceed 27 for $k \geq 4$.

Next, since $2^6 \equiv 1 \pmod{9}$ and $2006 \equiv 2 \pmod{6}$, we have

$$2^{2006} \equiv 2^2 \equiv 4 \pmod{9}.$$

We easily find that $f_k(2^{2006}) \equiv 7 \pmod{9}$ for odd k and $f_k(2^{2006}) \equiv 4 \pmod{9}$ for even k . Thus, the sum of digits of $f_{2005}(2^{2006})$ can only be 7, 16, 25. This yields $f_{2006}(2^{2006}) = 49, 256, 625$ and hence $f_{2007}(2^{2006}) = 169$.

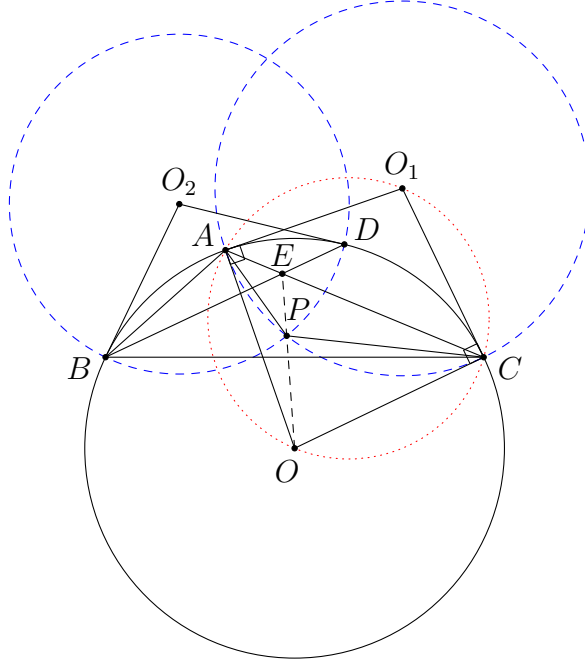
3. We only work on the configuration as shown since the other cases are similar.
We have

$$\angle CPA = \angle BAP + \angle CBA + \angle PCB = 90^\circ + \angle CBA.$$

Let O_1 be the centre of (APC) . Then we find that

$$\angle AO_1C = 360^\circ - 2\angle CPA = 180^\circ - 2\angle CBA = 180^\circ - 2\angle COA.$$

This shows A, O, C, O_1 are concyclic. As O, O_1 lie on the perpendicular bisector of AC , O_1 is just the intersection of the tangents at A and C to $(ABCD)$. Similarly, let O_2 be the intersection of the tangents at B and D to $(ABCD)$. Then O_2 is the centre of (BPD) .



Clearly, P lies on the radical axis of (APC) and (BPD) . Secondly, since we have $EA \times EC = EB \times ED$, E also lies on this radical axis. Lastly, the powers of O with respect to the two circles are OA^2 and OB^2 respectively as OA and OB are the tangents. Since $OA = OB$, these powers are equal so that O lies on the radical axis. Therefore, P, E, O are collinear.

4. We say that a sequence $\{b_n\}$ of positive real numbers is *good* if there exists a positive constant N such that

$$b_1 + b_2 + \cdots + b_n < Nb_{n+1} \tag{1}$$

for any $n \in \mathbb{Z}^+$.

We first prove the following equivalent conditions for a sequence to be good.

Claim. A sequence $\{b_n\}$ of positive real numbers is good if and only if the following conditions hold.

- (i) There exists a constant $r > 0$ such that $b_{n+1} > rb_n$ for any $n \in \mathbb{Z}^+$; and
- (ii) there exists a positive integer d such that $b_{n+d} > 2b_n$ for any $n \in \mathbb{Z}^+$.

Proof. Suppose $\{b_n\}$ is good. Then we have $Nb_{n+1} > b_n$. So we can take $r = \frac{1}{N}$ for (i). Next, we take d to be any integer larger than $2N^2 + 1$. Note that $Nb_{n+j} > b_1 + b_2 + \cdots + b_{n+j-1} > b_n$ for any $j \in \mathbb{Z}^+$. It follows that

$$\begin{aligned}
 b_{n+d} &> \frac{1}{N}(b_1 + b_2 + \cdots + b_{n+d-1}) \\
 &> \frac{1}{N}(b_{n+1} + b_{n+2} + \cdots + b_{n+d-1}) \\
 &> \frac{1}{N^2}(b_n + b_n + \cdots + b_n) \\
 &= \frac{d-1}{N^2}b_n \\
 &> 2b_n.
 \end{aligned}$$

This proves (ii).

Conversely, suppose (i) and (ii) hold. For any $n \in \mathbb{Z}^+$, we have

$$\begin{aligned}
 b_1 + b_2 + \cdots + b_n &= \sum_{j=1}^d (b_{n+1-j} + b_{n+1-j-d} + b_{n+1-j-2d} + \cdots) \\
 &< \sum_{j=1}^d \left(b_{n+1-j} + \frac{1}{2}b_{n+1-j} + \frac{1}{2^2}b_{n+1-j} + \cdots \right) \\
 &< \sum_{j=1}^d 2b_{n+1-j} \\
 &< 2 \left(\sum_{j=1}^d \frac{1}{r^j} \right) b_{n+1}.
 \end{aligned}$$

So we can take $N = 2 \left(\sum_{j=1}^d \frac{1}{r^j} \right)$ so that (1) holds. □

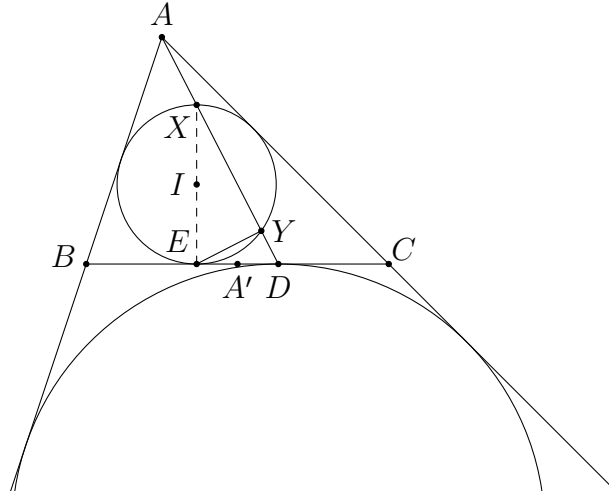
We go back to the original problem. Note that the sequence $\{a_n^2\}$ is good by the given condition. By the claim, there exists $r > 0$ and $d \in \mathbb{Z}^+$ such that $a_{n+1}^2 > ra_n^2$ and $a_{n+d}^2 > 2a_n^2$ for any $n \in \mathbb{Z}^+$. This implies $a_{n+1} > \sqrt{r}a_n$ and

$$a_{n+2d} > \sqrt{2}a_{n+d} > 2a_n.$$

Therefore, the sequence $\{a_n\}$ is good by the claim. This is exactly our goal.

Year 2007

1. (i) Note that $AB+BD$ is the semiperimeter of $\triangle ABC$, and so D is the contact point of the A -excircle of $\triangle ABC$ and BC . Since A is a centre of homothety between the incircle and the A -excircle, X and D are corresponding points under this homothety. Therefore, the tangent at X to the incircle is parallel to BC . This implies XE is a diameter of the incircle. It follows that $\angle XYE = 90^\circ$, and hence $EY \perp AD$.



- (ii) Note that I and A' are midpoints of EX and ED respectively. By the midpoint theorem, $XD = 2IA'$.
2. Yes. For example, we can take $f(x) = x^{2007} - x^{2006} + 1$.

It suffices to show that $(m, f^k(m)) = 1$ for any positive integer k , since we can replace m by any $f^j(n)$. Consider any prime p dividing m . Then we have $f(m) \equiv 0 - 0 + 1 = 1 \pmod{p}$. Whenever $f^i(m) \equiv 1 \pmod{p}$, we have

$$f^{i+1}(m) \equiv 1 - 1 + 1 \equiv 1 \pmod{p}.$$

Thus, $f^k(m) \equiv 1 \pmod{p}$ for any $k \geq 1$ by induction. This shows $p \nmid f^k(m)$, and hence $(m, f^k(m)) = 1$. This completes the proof.

3. Suppose on the contrary that no club contains at least 11 male and 11 female members. We shall count the number of triples (M, F, C) where M is a male student, F is a female student and C is a club joined by these two students.

Firstly, for each of the 2007 choices of M and each of the 2007 choices of F , the condition shows that there is at least one club C for which (M, F, C) should be counted. This gives at least 2007^2 triples.

Secondly, we count according to the clubs. We label the clubs as $C_1, C_2, \dots, C_k$. For each $i = 1, 2, \dots, k$, let m_i and f_i be the numbers of male students and female students in the club C_i respectively. Then the number of triples (M, F, C) is

$$\sum_{i=1}^k m_i f_i.$$

From the assumption, we can divide these clubs into two groups. The first group consists of those clubs with at most 10 male members (say $C_1, C_2, \dots, C_\ell$). The second group consists of those clubs with at least 11 male members, and hence at most 10 female members (say $C_{\ell+1}, C_{\ell+2}, \dots, C_k$). Then

$$\begin{aligned} \sum_{i=1}^k m_i f_i &= \sum_{i=1}^{\ell} m_i f_i + \sum_{i=\ell+1}^k m_i f_i \\ &\leq 10 \sum_{i=1}^{\ell} f_i + 10 \sum_{i=\ell+1}^k m_i \\ &\leq 10 \sum_{i=1}^k f_i + 10 \sum_{i=1}^k m_i. \end{aligned}$$

Note that the sum over f_i counts the total number of female members in all clubs. By another simple double counting, this is equal to the total number of clubs joined by all the female students. Therefore, we have

$$\sum_{i=1}^k f_i \leq 100 \times 2007.$$

The same upper bound holds for the sum over m_i . Combining all these, we obtain

$$2007^2 \leq 10 \times 100 \times 2007 + 10 \times 100 \times 2007 = 2000 \times 2007.$$

This is a contradiction. Hence, there must be a club with at least 11 male and 11 female members.

4. Yes, it exists.

There are finitely many fractions $\frac{a}{b}$ of lowest term such that $a \geq 1$, $1 \leq b \leq 2007$ and $\frac{a}{b} < \sqrt{2}$. Let $\frac{n}{m}$ be the largest such fraction. If $\left\lceil \frac{nk}{m} \right\rceil \neq \lceil \sqrt{2}k \rceil$ for some $k = 1, 2, \dots, 2007$, then $t = \lceil \sqrt{2}k \rceil$ is an integer satisfying $\frac{nk}{m} < t < \sqrt{2}k$ since $\frac{n}{m} < \sqrt{2}$ and $\sqrt{2}k \notin \mathbb{Z}$. It follows that $\frac{n}{m} < \frac{t}{k} < \sqrt{2}$, which contradicts the choice of $\frac{n}{m}$. Therefore, we must have $\left\lceil \frac{nk}{m} \right\rceil = \lceil \sqrt{2}k \rceil$ for $1 \leq k \leq 2007$.

Year 2008

1. (a) Recall the fact that $b - c \mid f(b) - f(c)$ for any integers b and c . Putting $b = a_{i+1}$ and $c = a_i$, we immediately obtain

$$a_{i+1} - a_i \mid f(a_{i+1}) - f(a_i) = a_{i+2} - a_{i+1}.$$

By induction on j , $a_{i+1} - a_i \mid a_{j+1} - a_j$ for any $j > i$.

- (b) Suppose on the contrary that $a_{2008} = 0$. Note that $a_{2009} - a_{2008} = f(0) = a_2 - a_1$. By part (a), since $a_2 - a_1 \mid a_{k+1} - a_k \mid a_{2009} - a_{2008}$ for $k = 1, 2, \dots, 2007$, each $a_{k+1} - a_k$ must be $\pm f(0)$. Now, we have

$$0 = a_{2008} - a_1 = \sum_{k=1}^{2007} (a_{k+1} - a_k) = mf(0),$$

where $m \in \{-2007, -2005, \dots, -1, 1, 3, \dots, 2007\}$. This implies $f(0) = 0$, which contradicts the assumption that $c_0 \neq 0$. Therefore, $a_{2008} \neq 0$.

2. If $p^2 \mid n$ for some prime p , then $p \mid \varphi(p^2) \mid \varphi(n)$. This implies $p \nmid \varphi(n)\sigma(n) + 1$, and hence $n \nmid \varphi(n)\sigma(n) + 1$. Therefore, n must be squarefree.

If $n = pq$ for some distinct primes p and q , then

$$\varphi(pq)\sigma(pq) + 1 = (p-1)(q-1)(p+1)(q+1) + 1 = p^2q^2 - p^2 - q^2 + 2.$$

This is a multiple of pq if and only if $pq \mid p^2 + q^2 - 2$. We now prove that $ab \mid a^2 + b^2 - 2$ has no solution in distinct integers $a, b > 1$ using Vieta's jumping.

Suppose on the contrary that the divisibility is solvable. Let (a, b) be a solution such that $a > b > 1$ and b is the smallest possible. Let $a^2 + b^2 - 2 = kab$ for some positive integer k . Rewrite this equation as

$$a^2 - kba + (b^2 - 2) = 0.$$

This is a quadratic equation in a . Let a' be another solution to this equation. Since $a + a' = kb$, a' is an integer. Since $aa' = b^2 - 2 > 0$, a' is positive. Also, we have

$$a' = \frac{b^2 - 2}{a} < \frac{b^2}{a} < b < a.$$

This shows $(x, y) = (b, a')$ is another pair of solution to $xy \mid x^2 + y^2 - 2$ such that $x > y > 0$. As $a' < b$, by the minimality of b , we must have $a' = 1$. But then the divisibility becomes $b \mid 1 + b^2 - 2$, i.e. $b \mid 1$. This is impossible since $b > 1$. Thus, there is no solution.

It follows that n must have at least 3 distinct prime divisors.

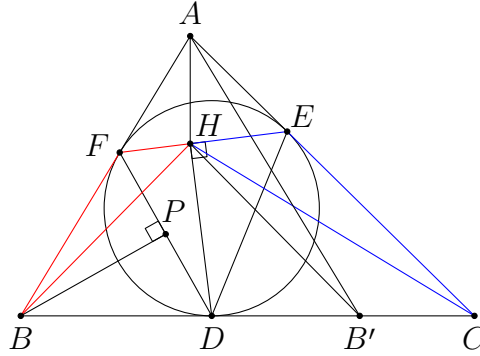
3. Let P be the midpoint of DF . Then $\angle FPB = 90^\circ$. Since $\angle BFP = \angle DEH$, we have $\triangle BPF \sim \triangle DHE$. This implies $\frac{BF}{PF} = \frac{DE}{HE}$, which yields

$$BF \times HE = DE \times PF = \frac{1}{2}DE \times DF.$$

By symmetry,

$$CE \times HF = \frac{1}{2}DE \times DF = BF \times HE.$$

Together with $\angle BFH = 90^\circ + \frac{A}{2} = \angle CEH$, we obtain $\triangle BFH \sim \triangle CEH$. It follows that $\angle FBH = \angle ECH$.



Let B' be the reflection of B in the line AH . Note that B' lies on BC since $AH \perp BC$. Also, $B' \neq C$ as $\triangle ABC$ is scalene. Therefore, we find that

$$\angle AB'H = \angle ABH = \angle ACH.$$

This implies A, H, B', C are concyclic. Thus,

$$\angle HBC = \angle HB'B = \angle HAC = 90^\circ - C,$$

and hence $BH \perp AC$. So H is the orthocentre of $\triangle ABC$.

4. The smallest possible value of N is 2008.

Let $\Gamma_1, \Gamma_2, \dots, \Gamma_{2008}$ be the circles, and let $P_1, P_2, \dots, P_N$ be all the intersection points. For each P_i , suppose it lies on exactly n_i circles. Define $f(P_i, \Gamma_j) = \frac{1}{n_i}$ if P_i lies on Γ_j , and $f(P_i, \Gamma_j) = 0$ if P_i does not lie on Γ_j .

Claim. For each circle Γ_j , we have $\sum_{i=1}^N f(P_i, \Gamma_j) \geq 1$.

Proof. Let P_k be one of the intersection points on Γ_j such that n_k is the largest possible. For each of the $n_k - 1$ other circles passing through P_k , it intersects

Γ_j at another point Q_i different from P_k . Also, all these points Q_i are distinct since the circles are of the same size (equivalently, no three circles meet at the same two points). It follows that

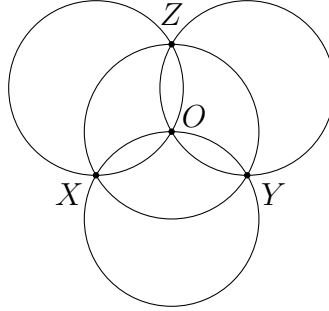
$$\sum_{i=1}^N f(P_i, \Gamma_j) \geq f(P_k, \Gamma_j) + \sum_{i=1}^{n_k-1} f(Q_i, \Gamma_j) \geq \frac{1}{n_k} + (n_k - 1) \cdot \frac{1}{n_k} = 1.$$

□

Now, since $\sum_{j=1}^{2008} f(P_i, \Gamma_j) = n_i \cdot \frac{1}{n_i} = 1$ for any intersection point P_i , we find that

$$N = \sum_{i=1}^N \sum_{j=1}^{2008} f(P_i, \Gamma_j) = \sum_{j=1}^{2008} \sum_{i=1}^N f(P_i, \Gamma_j) \geq \sum_{j=1}^{2008} 1 = 2008.$$

The bound 2008 can be attained. Consider an equilateral $\triangle XYZ$ with centre O . Then the four circles (XYZ) , (OXY) , (OYZ) , (OZX) each forms 3 intersection points with the other circles. Therefore, we can form 2008 circles by using 502 such groups. This gives exactly 2008 intersection points.



Year 2009

1. The smallest possible integer is 269.

For $n > 1$, we rewrite the recurrence relation as follows.

$$\begin{aligned} n(a_{n+1} - a_n + 3) &= a_{n+1} + a_n + 3 \\ \Rightarrow (n-1)a_{n+1} &= (n+1)a_n - 3(n-1) \\ \Rightarrow \frac{a_{n+1}}{n(n+1)} &= \frac{a_n}{(n-1)n} - \frac{3}{n(n+1)} \end{aligned}$$

Let $b_n = \frac{a_n}{(n-1)n}$. This gives $b_{n+1} = b_n - \frac{3}{n(n+1)}$. Therefore, we easily obtain

$$\begin{aligned} b_n &= b_{n-1} - \frac{3}{(n-1)n} \\ &= b_{n-2} - \frac{3}{(n-2)(n-1)} - \frac{3}{(n-1)n} \\ &= \dots \\ &= b_2 - \sum_{k=2}^{n-1} \frac{3}{k(k+1)} \\ &= b_2 - 3 \sum_{k=2}^{n-1} \left(\frac{1}{k} - \frac{1}{k+1} \right) \\ &= b_2 - 3 \left(\frac{1}{2} - \frac{1}{n} \right). \end{aligned}$$

Let $a_2 = 2m + 1$. It follows that

$$a_n = (n-1)n \left(\frac{a_2}{2} - 3 \left(\frac{1}{2} - \frac{1}{n} \right) \right) = (n-1)((m-1)n + 3).$$

It is given that $2010 \mid a_{2009} = 2008(2009(m-1) + 3)$. This can be reduced to $1005 \mid 2009(m-1) + 3$. As $2009 \equiv -1 \pmod{1005}$, this yields $m \equiv 4 \pmod{1005}$. Let $m = 1005t + 4$. Then

$$a_n = (n-1)((1005t + 3)n + 3) = 3(n-1)((335t + 1)n + 1).$$

Note that $2010 \mid a_n$ if and only if $670 \mid (n-1)((335t + 1)n + 1)$. It suffices to consider odd n . Therefore, this becomes $670 \mid (n-1)(n+1)$.

Now, $670 = 2 \times 5 \times 67$. Therefore, $n \equiv \pm 1 \pmod{67}$. It is routine to check that $670 \nmid (n-1)(n+1)$ when $n = 133, 135, 267$. The next smallest possible n is $n = 269$, where $670 \mid 268 \times 270$. So this is the answer.

2. It suffices to show that the case $n = 13$ is impossible since we can remove extra points. For the j th point, let r_j, y_j, g_j be the numbers of lines having this point as an endpoint which are in red, yellow, and green respectively.

For any $\triangle XYZ$, WLOG assume XY and XZ are red. Then this triangle corresponds to two red lines from X , and this correspondence is one-to-one. It follows that the number of triangles is

$$\sum_{j=1}^{13} \left[\binom{r_j}{2} + \binom{y_j}{2} + \binom{g_j}{2} \right].$$

But then it is clear that there are $\binom{13}{3} = 286$ triangles. By the pigeonhole principle, WLOG assume

$$\binom{r_1}{2} + \binom{y_1}{2} + \binom{g_1}{2} \geq \frac{286}{13} = 22.$$

This gives $r_1(r_1 - 1) + y_1(y_1 - 1) + g_1(g_1 - 1) \geq 44$. Since $r_1 + y_1 + g_1 = 12$, this implies $r_1^2 + y_1^2 + g_1^2 \geq 56$.

We claim that one of r_1, y_1, g_1 is at least 6. Suppose on the contrary that $r_1, y_1, g_1 \leq 5$. As $r_1 + y_1 + g_1 = 12$, it suffices to check $(r_1, y_1, g_1) = (5, 5, 2), (5, 4, 3), (4, 4, 4)$. In all cases, $r_1^2 + y_1^2 + g_1^2 \geq 56$ does not hold.

WLOG assume $r_1 \geq 6$. Suppose AB, AC, AD, AE, AF, AG are red. By assumption, none of the lines formed by B, C, D, E, F, G is red. Since $R(3, 3) = 6$, there must be a yellow or green triangle among these 6 points, contradiction. This proves $n < 13$.

3. (i) This is an immediate consequence of Casey's theorem. We provide an elementary proof as follows assuming the result in part (ii) (which will be proved later).

Let $a = BC$, $b = CA$ and $c = AB$. By part (ii), we have $BM = BC = a$. Then we have

$$\begin{aligned} AN &= AM = c - a, \\ CN &= b - AN = a + b - c, \\ DM &= AD - AM = \frac{b^2}{c} - c + a. \end{aligned}$$

It follows that

$$\begin{aligned} & BD \cdot CN + BC \cdot DM - CD \cdot BM \\ &= \frac{a^2}{c}(a + b - c) + a \left(\frac{b^2}{c} - c + a \right) - \frac{ab}{c} \cdot a \\ &= \frac{a}{c}(a^2 + ab - ac + b^2 - c^2 + ac - ab) = 0 \end{aligned}$$

using Pythagoras' theorem. This proves the equality.

This is the same as $(m+1)^2 \equiv 5 \pmod{113}$. However, we check that

$$\left(\frac{5}{113}\right) = \left(\frac{113}{5}\right) = \left(\frac{3}{5}\right) = -1,$$

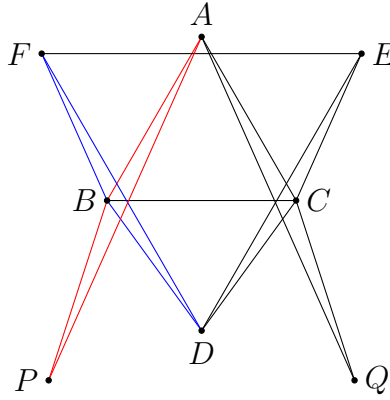
which implies 5 is a quadratic nonresidue modulo 113. This is a contradiction. Therefore, there is no solution.

Thus, the only solutions are $(m, n) = (1, 0), (1, 113^{114})$.

Year 2010

1. Let D, E, F be the centres of the regular n -gons respectively having BC, CA, AB as one of the sides. Let PB, BC, CQ be three consecutive sides of the regular n -gon with centre D .

Note that $\triangle BPD \sim \triangle BAF$ since all regular n -gons are similar. By spiral similarity, we have $\triangle BPA \sim \triangle BDF$. This implies $\frac{PA}{DF} = \frac{BA}{BF} = 2 \sin \frac{\pi}{n}$. By symmetry, we also have $\frac{QA}{DE} = 2 \sin \frac{\pi}{n}$. This shows $\frac{PA}{DF} = \frac{QA}{DE}$.



If $n = 3$, then $P = Q$. Trivially, $PA = QA$. This yields $DF = DE$. By symmetry, $\triangle DEF$ is equilateral. Indeed, this is called the outer Napoleon triangle of $\triangle ABC$.

If $n > 3$, we show that $\triangle DEF$ cannot be equilateral. Suppose on the contrary that $\triangle DEF$ is equilateral. Then $DF = DE$. Using $\frac{PA}{DF} = \frac{QA}{DE}$, we obtain $PA = QA$. Note that P and Q are distinct points. Therefore, A lies on the perpendicular bisector of PQ , which is also the perpendicular bisector of BC . Thus, $AB = AC$. This contradicts the assumption that $\triangle ABC$ is scalene.

2. The answer is $2 \cdot 3^n - 2^n$.

Let $y_j = x_{2j-1} + x_{2j}$ for $j = 1, 2, \dots, n$. Every desired sequence $\{x_j\}$ corresponds to a sequence $\{y_j\}$ such that $y_j = -2, 0, 2$ for each j , and $|y_k + y_{k+1} + \dots + y_m| \leq 2$ for any $1 \leq k \leq m \leq n$. Suppose there are r terms of $\{y_j\}$ which are nonzero, say $y_{i_1}, y_{i_2}, \dots, y_{i_r}$ where $i_1 < i_2 < \dots < i_r$. Since

$$|y_{i_1} + y_{i_1+1} + \dots + y_{i_2}| = |y_{i_1} + y_{i_2}| \leq 2,$$

we must have $y_{i_2} = -y_{i_1}$. Similarly, $y_{i_3} = -y_{i_2} = y_{i_1}$, and inductively $y_{i_{2j-1}} = y_{i_1}$ and $y_{i_{2j}} = -y_{i_1}$ for any integer j . In other words, the sequence $\{y_j\}$ only depends on the indices $i_1, i_2, \dots, i_r$ and the term y_{i_1} .

Conversely, for any such sequence $\{y_j\}$, consider any $1 \leq k \leq m \leq n$. Let i_s be the smallest index such that $k \leq i_s$, and let i_t be the largest index such that $i_t \leq m$. Then

$$y_k + y_{k+1} + \dots + y_m = y_{i_s} + y_{i_s+1} + \dots + y_{i_t} = \pm(2 - 2 + 2 - 2 + \dots) = -2, 0, 2.$$

So the condition is satisfied.

Now, for each sequence $\{y_j\}$ satisfying the condition, we can construct the sequence $\{x_j\}$ as follows. For any index j of the form i_u , the terms x_{2j-1} and x_{2j} are uniquely determined. Indeed, if $y_{i_u} = \pm 2$, then $x_{2i_u-1} = x_{2i_u} = \pm 1$. For any other index j , there are two choices for x_{2j-1} and x_{2j} , since exactly one of them is 1 and the other is -1. Since there are $\binom{n}{r}$ ways to choose the indices $i_1, i_2, \dots, i_r$ and two ways to choose y_{i_1} if $r \geq 1$, the total number of sequences is

$$\binom{n}{0}2^n + \sum_{r=1}^n 2\binom{n}{r} \cdot 2^{n-r} = 2^n + 2 \sum_{r=1}^n \binom{n}{n-r} 2^{n-r} = 2^n + 2 \sum_{d=0}^{n-1} \binom{n}{d} 2^d$$

using the change of variable $d = n - r$. By the binomial theorem, this is equal to

$$2^n + 2 \sum_{d=0}^n \binom{n}{d} 2^d - 2^{n+1} = 2^n + 2(2+1)^n - 2^{n+1} = 2 \cdot 3^n - 2^n.$$

3. For each integer j relatively prime to n , let r_j be the remainder when aj is divided by n . Then we have

$$\begin{aligned} a^{\varphi(n)} &= \prod_{\substack{j=1 \\ (j,n)=1}}^n \frac{aj}{j} = \prod_{\substack{j=1 \\ (j,n)=1}}^n \frac{1}{j} \left(\left[\frac{aj}{n} \right] n + r_j \right) \\ &= \prod_{\substack{j=1 \\ (j,n)=1}}^n \frac{r_j}{j} \left(1 + \frac{n}{r_j} \left[\frac{aj}{n} \right] \right) = \prod_{\substack{j=1 \\ (j,n)=1}}^n \left(1 + \frac{n}{r_j} \left[\frac{aj}{n} \right] \right) \end{aligned}$$

since $\{r_1, r_2, \dots, r_{\varphi(n)}\}$ is a reduced set of residues modulo n . Expanding the product and taking modulo n^2 , we obtain

$$a^{\varphi(n)} \equiv 1 + n \sum_{\substack{j=1 \\ (j,n)=1}}^n \frac{1}{r_j} \left[\frac{aj}{n} \right] \equiv 1 + n \sum_{\substack{j=1 \\ (j,n)=1}}^n \frac{1}{aj} \left[\frac{aj}{n} \right] \pmod{n^2}.$$

This implies

$$\frac{a^{\varphi(n)} - 1}{n} \equiv \sum_{\substack{j=1 \\ (j,n)=1}}^n \frac{1}{aj} \left[\frac{aj}{n} \right] \pmod{n}.$$

4. (Iran TST 1996) Since the expressions are homogeneous, WLOG we may assume $ab + bc + ca = 1$. We need to minimize

$$f(a, b, c) = \frac{1}{(a+b)^2} + \frac{1}{(b+c)^2} + \frac{1}{(c+a)^2}.$$

WLOG assume $a \geq b \geq c$. Let t be the positive root to $t^2 + 2ct = 1$, so that $(t)(t) + (t)(c) + (c)(t) = 1$, which means the triple (t, t, c) also satisfies the constraint.

Claim. We have $f(a, b, c) \geq f(t, t, c)$.

Proof. Firstly, we have

$$f(t, t, c) = \frac{1}{4t^2} + \frac{2}{(t+c)^2} = \frac{1}{4t^2} + \frac{2}{t^2 + 2ct + c^2}.$$

For the latter term, observe that $t^2 + 2ct = 1 = ab + bc + ca$. Therefore, we obtain

$$\frac{2}{t^2 + 2ct + c^2} = \frac{2}{ab + bc + ca + c^2} = \frac{2}{(c+a)(c+b)}.$$

For the former term, we claim that $t^2 \geq ab$. Suppose on the contrary that $t^2 < ab$. Then we also have $a + b \geq 2\sqrt{ab} > 2t$. But this implies

$$ab + bc + ca = ab + c(a+b) > t^2 + 2ct,$$

which is a contradiction. Therefore, $t^2 \geq ab$, and hence

$$\frac{1}{4t^2} \leq \frac{1}{4ab}.$$

Now, it suffices to show $f(a, b, c) \geq \frac{1}{4ab} + \frac{2}{(c+a)(c+b)}$. Indeed,

$$\begin{aligned} f(a, b, c) &\geq \frac{1}{4ab} + \frac{2}{(c+a)(c+b)} \\ \Leftrightarrow \frac{1}{(b+c)^2} + \frac{1}{(c+a)^2} - \frac{2}{(c+a)(c+b)} &\geq \frac{1}{4ab} - \frac{1}{(a+b)^2} \\ \Leftrightarrow \frac{((c+a) - (c+b))^2}{(c+a)^2(c+b)^2} &\geq \frac{(a+b)^2 - 4ab}{4ab(a+b)^2} \\ \Leftrightarrow \frac{(a-b)^2}{(c+a)^2(c+b)^2} &\geq \frac{(a-b)^2}{4ab(a+b)^2}. \end{aligned}$$

This is true since

$$(c+a)^2(c+b)^2 \leq (b+a)^2(b+b)^2 = 4b^2(a+b)^2 \leq 4ab(a+b)^2$$

using the assumption $a \geq b \geq c$. This proves the claim. $\square$

By the claim, it remains to prove $f(t, t, c) \geq \frac{9}{4}$ for any positive real numbers t and c satisfying $t^2 + 2ct = 1$. Now, using $c = \frac{1-t^2}{2t}$,

$$\begin{aligned}
& f(t, t, c) \geq \frac{9}{4} \\
\Leftrightarrow & \frac{1}{4t^2} + \frac{2}{(t+c)^2} \geq \frac{9}{4} \\
\Leftrightarrow & \frac{1}{4t^2} + \frac{8t^2}{(1+t^2)^2} \geq \frac{9}{4} \\
\Leftrightarrow & \frac{-9t^6 + 15t^4 - 7t^2 + 1}{4t^2(1+t^2)^2} \geq 0 \\
\Leftrightarrow & \frac{(1-t^2)(1-3t^2)^2}{4t^2(1+t^2)^2} \geq 0.
\end{aligned}$$

As $1 = t^2 + 2ct > t^2$, this inequality is true. Equality holds when $t = \frac{1}{3}$, i.e. $a = b = c$. (Note that if we allow at most one a, b, c to be 0, equality also holds when $t = 1$, i.e. $a = b$ and $c = 0$ up to permutation.)

Year 2011

1. We rewrite the recurrence relation as follows.

$$\begin{aligned}
 x_{n+1} &= \sqrt{5}x_n + 2\sqrt{x_n^2 + 1} \\
 \Rightarrow (x_{n+1} - \sqrt{5}x_n)^2 &= 4(x_n^2 + 1) \\
 \Rightarrow x_n^2 - 2\sqrt{5}x_{n+1}x_n + (x_{n+1}^2 - 4) &= 0.
 \end{aligned}$$

This is a quadratic equation in x_n . We use the quadratic formula to obtain

$$x_n = \sqrt{5}x_{n+1} \pm \sqrt{5x_{n+1}^2 - (x_{n+1}^2 - 4)} = \sqrt{5}x_{n+1} \pm 2\sqrt{x_{n+1}^2 + 1}.$$

Using the given relation, it is clear that $x_{n+1} > x_n$. Therefore,

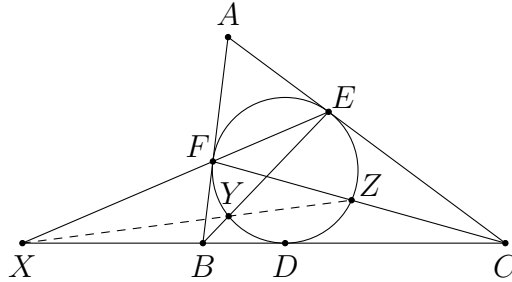
$$x_n = \sqrt{5}x_{n+1} - 2\sqrt{x_{n+1}^2 + 1}.$$

Together with $x_{n+2} = \sqrt{5}x_{n+1} + 2\sqrt{x_{n+1}^2 + 1}$, we obtain

$$x_{n+2} + x_n = 2\sqrt{5}x_{n+1}.$$

As $x_n, x_{n+1}, x_{n+2} > 0$, this relation suggests they cannot be all rational. Thus, there must be an irrational number in each of the groups $(x_{3k+1}, x_{3k+2}, x_{3k+3})$ where $k = 0, 1, \dots, 669$. That means there are at least 670 irrational numbers.

2. Let $X' = EF \cap YZ$. Apply Pascal's theorem to the points $EEFFZY$ on the incircle. Then $C = EE \cap FZ$, $X' = EF \cap ZY$ and $B = FF \cap YE$ are collinear. Hence, $X' = X$, and X, Y, Z are collinear.



3. (IMO Shortlist 1996 N3 modified)

(a) Observe that $f(k) = k^2 - (k+1)^2 - (k+2)^2 + (k+3)^2 = 4$ for any integer k .

- If $c = b + 4m$ for some $m \in \mathbb{Z}$, then $c = b + \sum_{k=1}^m f(4k-3)$.
- If $c = b + 4m + 1$ for some $m \in \mathbb{Z}$, then $c = b + 1^2 + \sum_{k=1}^m f(4k-2)$.
- If $c = b + 4m + 2$ for some $m \in \mathbb{Z}$, then $c = b - 1^2 - 2^2 - 3^2 + 4^2 + \sum_{k=1}^m f(4k+1)$.

- If $c = b + 4m + 3$ for some $m \in \mathbb{Z}$, then $c = b - 1^2 + 2^2 + \sum_{k=1}^m f(4k - 1)$.

Therefore, in any case we can write c as $b \pm 1^2 \pm 2^2 \pm \dots \pm n^2$ for some n and some choices of the signs. This clearly solves the problem.

(b) The answer is 19.

Firstly, since $1^2 + 2^2 + \dots + 17^2 = \frac{17 \cdot 18 \cdot 35}{6} = 1785 < 2012$, we have $n \geq 18$.

Secondly, when $n = 18$, $\pm 1^2 \pm 2^2 \pm \dots \pm 18^2 \equiv 1 + 0 + \dots + 1 + 0 \equiv 1 \pmod{2}$. This shows b and c cannot have the same parity, which is a contradiction.

Lastly, it remains to provide a construction for $n = 19$. Since

$$1^2 + 2^2 + \dots + 19^2 = \frac{19 \cdot 20 \cdot 39}{6} = 2470 = 2012 + 2 \times 229 = 2012 + 2(2^2 + 15^2),$$

we can change the signs of 2^2 and 15^2 to negative, so that the new sum becomes 2012. This gives a corresponding quadratic sequence that satisfies the requirement.

4. The maximum number of acute-angled triangles is $\frac{n(n-1)(n+1)}{24}$ if n is odd and is $\frac{n(n-2)(n+2)}{24}$ if n is even.

We first count the number N_1 of non-acute triangles with A_1 as one vertex such that $\angle A_1 < 90^\circ$. Let ℓ be the diameter of the circle which passes through A_1 . Then for points B, C on the circle, $\triangle A_1 BC$ is not acute with $\angle BA_1 C < 90^\circ$ if and only if B and C lie on the same side of ℓ (possibly on ℓ). Suppose there are a points on one side of ℓ and b points on the other side of ℓ other than A_1 . Note that the two sides share at most one common point which is the other intersection of ℓ and the circle. Therefore, we must have $a + b = n - 1$ or $a + b = n$. We find that

$$N_1 = \binom{a}{2} + \binom{b}{2}.$$

We have to minimize N_1 . Note that $b \geq n - 1 - a$ so that $\binom{b}{2} \geq \binom{n-1-a}{2}$. WLOG assume $a \geq n - 1 - a$. If n is odd, then $(a, n - 1 - a) > \left(\frac{n-1}{2}, \frac{n-1}{2}\right)$. Applying the majorization inequality, we obtain

$$N_1 \geq \binom{a}{2} + \binom{n-1-a}{2} \geq \binom{\frac{n-1}{2}}{2} + \binom{\frac{n-1}{2}}{2} = \frac{(n-1)(n-3)}{4}.$$

The same bound holds if we replace A_1 by any other vertex A_j . We sum up all the lower bounds. Note that each non-acute triangle is counted twice in

this sum (one corresponding to each acute angle). Therefore, the number of non-acute triangles is at least

$$n \cdot \frac{(n-1)(n-3)}{4} \cdot \frac{1}{2} = \frac{n(n-1)(n-3)}{8}.$$

Thus, the number of acute triangles is at most

$$\binom{n}{3} - \frac{n(n-1)(n-3)}{8} = \frac{n(n-1)(n+1)}{24}.$$

This can be attained when $A_1 A_2 \cdots A_n$ is a regular polygon (since in that case all inequalities used become equalities).

Similarly, if n is even, then $(a, n-1-a) > \left(\frac{n}{2}, \frac{n-2}{2}\right)$. Applying the majorization inequality, we obtain

$$N_1 \geq \binom{a}{2} + \binom{n-1-a}{2} \geq \binom{\frac{n}{2}}{2} + \binom{\frac{n-2}{2}}{2} = \frac{(n-2)^2}{4}.$$

Thus, the number of acute triangles is at most

$$\binom{n}{3} - n \cdot \frac{(n-2)^2}{4} \cdot \frac{1}{2} = \frac{n(n-2)(n+2)}{24}.$$

The regular polygon case does not yield equality. Indeed, we first consider a regular polygon $B_1 B_2 \cdots B_n$. Then we rotate half of the points, say $B_1, B_2, \dots, B_{\frac{n}{2}}$, anticlockwise about the centre of the circle with a sufficiently small angle θ to points $C_1, C_2, \dots, C_{\frac{n}{2}}$. Then $C_1 C_2 \cdots C_{\frac{n}{2}} B_{\frac{n}{2}+1} B_{\frac{n}{2}+2} \cdots B_n$ is one possible case which gives the upper bound $\frac{n(n-2)(n+2)}{24}$.

Year 2012

1. We say that a positive integer n is *excellent* if it is good, and we can find integers $c_1, c_2, \dots, c_m$ such that $\sum_{j=1}^m (-1)^{c_j} a_j = 0$. Let $p_1, p_2, \dots, p_k$ be distinct primes larger than 3. We prove by induction on k that the integer $6p_1p_2\cdots p_k$ is excellent.

For the base case $k = 0$, the positive divisors of 6 are 1, 2, 3, 6. Since we have $-1 - 2 + 3 + 6 = 6$ and $-1 - 2 - 3 + 6 = 0$, the integer 6 is excellent.

For the inductive step, assume $n = 6p_1p_2\cdots p_k$ is excellent. Let $a_1, a_2, \dots, a_m$ be all of its positive divisors, and let

$$\sum_{j=1}^m (-1)^{b_j} a_j = n \quad \text{and} \quad \sum_{j=1}^m (-1)^{c_j} a_j = 0.$$

Consider $n' = 6p_1p_2\cdots p_{k+1}$. Its positive divisors are of the form a_j and $a_j p_{k+1}$. Note that

$$\sum_{j=1}^m (-1)^{c_j} a_j + \sum_{j=1}^m (-1)^{b_j} (a_j p_{k+1}) = 0 + n p_{k+1} = n'.$$

Also, we have

$$\sum_{j=1}^m (-1)^{c_j} a_j + \sum_{j=1}^m (-1)^{c_j} (a_j p_{k+1}) = 0 + 0 \cdot p_{k+1} = 0.$$

Therefore, n' is excellent. By induction, our claim holds. In particular, we can take $k = 2011$ so that $6p_1p_2\cdots p_{2011}$ is excellent, and hence it is a good integer with exactly 2013 prime divisors.

2. The answer is 251.

Consider pairs of marked points with the same y -coordinates. If there exist two pairs of marked points with the same sum of x -coordinates and different y -coordinates, then the 4 points in these pairs are the vertices of an isosceles trapezoid.

Suppose there are $a_1, a_2, \dots, a_{101}$ marked points in the 101 horizontal lines respectively (this means there are a_1 marked points with y -coordinates equal to 1, etc.). For the horizontal line $y = 1$, suppose the marked points are $(x_1, 1), (x_2, 1), \dots, (x_{a_1}, 1)$ where $x_1 < x_2 < \cdots < x_{a_1}$. Since

$$x_1 + x_2 < x_1 + x_3 < \cdots < x_1 + x_{a_1} < x_2 + x_{a_1} < \cdots < x_{a_1-1} + x_{a_1},$$

there are at least $2a_1 - 3$ distinct sums formed by the x -coordinates. This only holds for $a_1 \geq 2$. For $a_1 = 0$ or $a_1 = 1$, clearly, no sum can be formed. Therefore, there are at least

$$f(a_1) + f(a_2) + \cdots + f(a_{101}) \tag{1}$$

sums where $f(n) = \max\{2n - 3, 0\}$. Each of these sums lies between $1 + 2 = 3$ and $100 + 101 = 201$. There are 199 possibilities. If (1) exceeds 199, then two of the sums are equal by the pigeonhole principle. Since the equal sums must correspond to distinct horizontal lines from the construction, we can find an isosceles trapezoid as suggested. Therefore, we may assume (1) is at most 199. Then

$$199 \geq f(a_1) + f(a_2) + \cdots + f(a_{101}) \geq (2a_1 - 3) + (2a_2 - 3) + \cdots + (2a_{101} - 3).$$

This implies $a_1 + a_2 + \cdots + a_{101} \leq 251$. In other words, the number of marked points is at most 251. This can be attained, for example, in the following construction.



It is not hard to see that there is no isosceles trapezoid (a rigorous proof is to show that the sums of x -coordinates of pairs of points in the same horizontal line are all different, and the same holds for y -coordinates by symmetry). There are

$$3 + 5 \times 49 + 3 = 251$$

points in total.

3. By the Cauchy-Schwarz inequality, we have

$$(1^2 a_1 + 2^2 a_2 + \cdots + k^2 a_k)(a_1^{-1} + a_2^{-1} + \cdots + a_k^{-1}) \geq (1 + 2 + \cdots + k)^2 = \frac{k^2(k+1)^2}{4}.$$

This implies

$$\frac{k}{a_1^{-1} + a_2^{-1} + \cdots + a_k^{-1}} \leq \frac{4}{k(k+1)^2} \sum_{j=1}^k j^2 a_j.$$

Also, using $\frac{1}{k^2} - \frac{1}{(k+1)^2} = \frac{2k+1}{k^2(k+1)^2} \geq \frac{2}{k(k+1)^2}$, we obtain

$$\begin{aligned}
\sum_{k=1}^n \frac{k}{a_1^{-1} + a_2^{-1} + \dots + a_k^{-1}} &\leq \sum_{k=1}^n \sum_{j=1}^k j^2 a_j \frac{4}{k(k+1)^2} \\
&\leq 2 \sum_{k=1}^n \sum_{j=1}^k j^2 a_j \left(\frac{1}{k^2} - \frac{1}{(k+1)^2} \right) \\
&= 2 \sum_{j=1}^n \sum_{k=j}^n j^2 a_j \left(\frac{1}{k^2} - \frac{1}{(k+1)^2} \right) \\
&= 2 \sum_{j=1}^n j^2 a_j \left(\frac{1}{j^2} - \frac{1}{(n+1)^2} \right) \\
&< 2 \sum_{j=1}^n a_j.
\end{aligned}$$

This proves the desired inequality.

Consider $a_j = \frac{1}{j}$ for each j . Then

$$\sum_{k=1}^n \frac{k}{a_1^{-1} + a_2^{-1} + \dots + a_k^{-1}} = \sum_{k=1}^n \frac{k}{1 + 2 + \dots + k} = \sum_{k=1}^n \frac{2}{k+1}$$

and $\sum_{k=1}^n a_k = \sum_{k=1}^n \frac{1}{k}$. For any c such that $0 < c < 2$, we have

$$c \sum_{k=1}^n a_k - \sum_{k=1}^n \frac{k}{a_1^{-1} + a_2^{-1} + \dots + a_k^{-1}} = c + (c-2) \sum_{k=2}^n \frac{1}{k} - \frac{2}{n+1} < 0$$

for sufficiently large n , since the harmonic series $\sum_{k=2}^{\infty} \frac{1}{k}$ diverges to positive infinity. This shows we cannot replace the number 2 by any smaller constant c .

4. Let N be the midpoint of $\widehat{BC}$ not containing A . Then A, I, N are collinear, and MN is a diameter of (ABC) . Since both ID and MN are perpendicular to BC , they are parallel. Therefore, we have

$$\angle AIP = \angle DPI = \angle IDP = \angle DIN = \angle MNI.$$

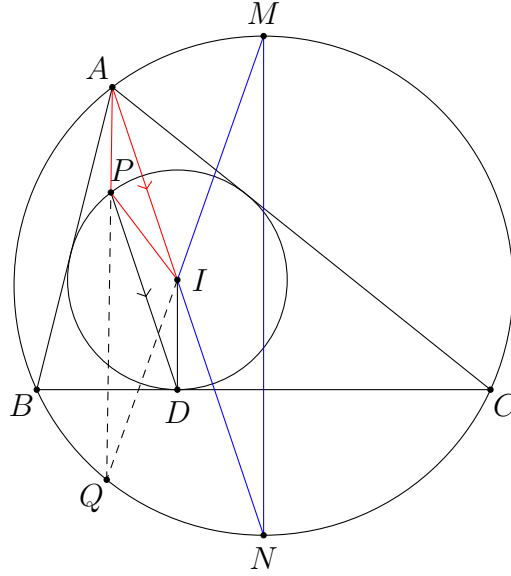
Let R and r be the circumradius and inradius of $\triangle ABC$. Considering the power of I with respect to (ABC) , we obtain

$$AI \times NI = R^2 - OI^2 = 2Rr = MN \times IP.$$

This implies $\frac{AI}{MN} = \frac{IP}{NI}$. Therefore, $\triangle AIP \sim \triangle MNI$. Let AP meet MI at Q . Then we have

$$\angle QAN = \angle PAI = \angle IMN = \angle QMN.$$

This implies Q, N, M, A are concyclic, which means Q lies on (ABC) as desired.



Year 2013

1. (IMO Shortlist 2004 A5 modified) By the power mean inequality, we have

$$\begin{aligned}
 S &:= \sqrt[4]{\frac{\sqrt{3}}{a} + 6\sqrt{3}b} + \sqrt[4]{\frac{\sqrt{3}}{b} + 6\sqrt{3}c} + \sqrt[4]{\frac{\sqrt{3}}{c} + 6\sqrt{3}a} \\
 &\leq 3 \sqrt[4]{\frac{1}{3} \left(\frac{\sqrt{3}}{a} + 6\sqrt{3}b + \frac{\sqrt{3}}{b} + 6\sqrt{3}c + \frac{\sqrt{3}}{c} + 6\sqrt{3}a \right)} \\
 &= 3^{\frac{7}{8}} \sqrt[4]{\frac{(ab + bc + ca) + 6abc(a + b + c)}{abc}}.
 \end{aligned}$$

Note that $ab + bc + ca = 1$ and

$$abc(a + b + c) = (ab)(bc) + (bc)(ca) + (ca)(ab) \leq \frac{1}{3}(ab + bc + ca)^2 = \frac{1}{3}.$$

Therefore, we have $S \leq 3^{\frac{7}{8}} \sqrt[4]{\frac{3}{abc}} = 3^{\frac{9}{8}}(abc)^{-\frac{1}{4}}$. It remains to prove $abc \leq 3^{-\frac{3}{2}}$. Indeed, by the AM-GM inequality, we have

$$1 = ab + bc + ca \geq 3\sqrt[3]{a^2b^2c^2}.$$

This clearly implies $abc \leq 3^{-\frac{3}{2}}$. This proves the desired inequality. Equality holds when $a = b = c = \frac{1}{\sqrt{3}}$.

2. The answer is $a = 1680$.

Since $ab \equiv a(-a) = -a^2 \pmod{a+b}$, we have $a+b \mid ab$ if and only if $a+b \mid a^2$. This shows $a+b$ can be any divisor of a^2 greater than a . Also, any such divisor corresponds to a unique positive integer b . Therefore, $M(a)$ is the number of divisors of a^2 greater than a . Since the positive divisors of a^2 less than a are in one-to-one correspondence to the divisors of a^2 greater than a , it remains to find a which has the maximum number of positive divisors.

For any $a = p_1^{c_1} p_2^{c_2} \cdots p_s^{c_s}$ where $p_1, p_2, \dots, p_s$ are distinct prime divisors of a , recall that $d(a^2) = (2c_1 + 1)(2c_2 + 1) \cdots (2c_s + 1)$. WLOG we may assume $c_1 \geq c_2 \geq \cdots \geq c_s$. Since $2 \times 3 \times 5 \times 7 \times 11 = 2310 > 2013$, we have $s \leq 4$.

- If $s = 4$, we must have $c_3 = c_4 = 1$ since $2^2 3^2 5^2 7 = 6300 > 2013$.
 - If $c_2 \geq 2$, we must have $c_1 = c_2 = 2$ since $2^3 3^2 5 \cdot 7 = 2520 > 2013$. When $a = p_1^2 p_2^2 p_3 p_4$, we have $d(a^2) = 5^2 \cdot 3^2 = 225$.
 - If $c_2 = 1$, we have $c_1 \leq 4$ since $2^5 3 \cdot 5 \cdot 7 = 3360 > 2013$. When $a = p_1^4 p_2 p_3 p_4$, we have $d(a^2) = 9 \cdot 3^3 = 243$.

- If $s = 3$, we must have $c_3 \leq 2$ since $2^3 3^3 5^3 = 27000 > 2013$.
 - If $c_3 = 2$, we must have $c_2 = 2$ since $2^3 3^3 5^2 = 5400 > 2013$. Then $c_1 \leq 3$ since $2^4 3^2 5^2 = 3600 > 2013$. When $a = p_1^3 p_2^2 p_3^2$, we have $d(a^2) = 7 \cdot 5^2 = 175$.
 - If $c_3 = 1$, we must have $c_2 \leq 3$ since $2^4 3^4 5 = 6480 > 2013$.
 - * If $c_2 = 3$, we have $c_1 = 3$ since $2^4 3^3 5 = 2160 > 2013$. When $a = p_1^3 p_2^3 p_3$, we have $d(a^2) = 7^2 \cdot 3 = 147$.
 - * If $c_2 = 2$, we have $c_1 \leq 5$ since $2^6 3^2 5 = 2880 > 2013$. When $a = p_1^5 p_2^2 p_3$, we have $d(a^2) = 11 \cdot 5 \cdot 3 = 165$.
 - * If $c_2 = 1$, we have $c_1 \leq 7$ since $2^8 3 \cdot 5 = 3840 > 2013$. When $a = p_1^7 p_2 p_3$, we have $d(a^2) = 15 \cdot 3^2 = 135$.
- If $s = 2$, we must have $c_2 \leq 4$ since $2^5 3^5 = 7776 > 2013$.
 - If $c_2 = 4$, we have $c_1 = 4$ since $2^5 3^4 = 2592 > 2013$. When $a = p_1^4 p_2^4$, we have $d(a^2) = 9^2 = 81$.
 - If $c_2 = 3$, we have $c_1 \leq 6$ since $2^7 3^3 = 3456 > 2013$. When $a = p_1^6 p_2^3$, we have $d(a^2) = 13 \cdot 7 = 91$.
 - If $c_2 = 2$, we have $c_1 \leq 7$ since $2^8 3^2 = 2304 > 2013$. When $a = p_1^7 p_2^2$, we have $d(a^2) = 15 \cdot 5 = 75$.
 - If $c_2 = 1$, we have $c_1 \leq 9$ since $2^{10} 3 = 3072 > 2013$. When $a = p_1^9 p_2$, we have $d(a^2) = 19 \cdot 3 = 57$.
- If $s \leq 1$, we have $c_1 \leq 10$ since $2^{11} = 2048 > 2013$. When $a = p_1^{10}$, we have $d(a^2) = 21$.

Therefore, a^2 has at most 243 positive divisors. This holds when $a = p_1^4 p_2 p_3 p_4$ for some distinct primes p_1, p_2, p_3, p_4 . As $3^4 2 \cdot 3 \cdot 5 = 2430 > 2013$, we must have $p_1 = 2$. Also, since $2^4 3 \cdot 5 \cdot 11 = 2640 > 2013$, the only possibility is $a = 2^4 3 \cdot 5 \cdot 7 = 1680$.

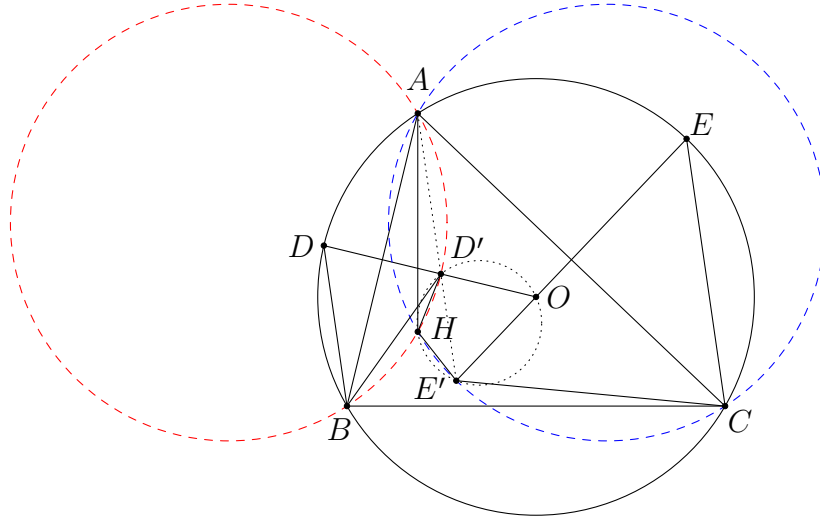
3. We claim that both statements are equivalent to $\angle A = 60^\circ$.

Firstly,

$$\begin{aligned}
 & A, D', E' \text{ are collinear} \\
 \Leftrightarrow & \angle BAD' + \angle CAE' = \angle BAC \\
 \Leftrightarrow & \angle BAD + \angle CAE = \angle BAC \\
 \Leftrightarrow & \frac{C}{2} + \frac{B}{2} = A \\
 \Leftrightarrow & 90^\circ - \frac{A}{2} = A \\
 \Leftrightarrow & A = 60^\circ.
 \end{aligned}$$

Secondly, it is well-known that the reflection of H in AB lies on (ABC) . Therefore, A, B, H, D' are concyclic. Also, note that D, D', O are collinear since all of them lie on the perpendicular bisector of AB . It follows that

$$\begin{aligned}
 \angle DD'H &= \angle DD'B + \angle BD'H \\
 &= (90^\circ - \angle ABD') + \angle BAH \\
 &= (90^\circ - \angle ABD) + \angle BAH \\
 &= \left(90^\circ - \frac{C}{2}\right) + 90^\circ - B \\
 &= A + \frac{C}{2}.
 \end{aligned}$$



Similarly, C, A, H, E' are concyclic, and we have

$$\begin{aligned}
 \angle OE'H &= \angle CE'H - \angle CE'E \\
 &= (180^\circ - \angle CAH) - (90^\circ - \angle ACE) \\
 &= (90^\circ + C) - \left(90^\circ - \frac{B}{2}\right) \\
 &= C + \frac{B}{2}.
 \end{aligned}$$

Now,

$$\begin{aligned}
 &O, H, D', E' \text{ are concyclic} \\
 \Leftrightarrow &\angle DD'H = \angle OE'H \\
 \Leftrightarrow &A + \frac{C}{2} = C + \frac{B}{2} \\
 \Leftrightarrow &A = 90^\circ - \frac{A}{2} \\
 \Leftrightarrow &A = 60^\circ.
 \end{aligned}$$

Therefore, the two statements are equivalent.

4. (CMO 1999 Problem 3 modified) The answer is $\frac{n(n-3)(n^2+6n-31)}{48}$.

For each k , let d_k be the number of games that player k wins. Note that

$$S := d_1 + d_2 + \cdots + d_n = \binom{n}{2} - n = \frac{n(n-3)}{2}.$$

Observe that for a set of four players, if one of them beats everybody else, then the set is not harmonic. Also, there cannot be two players beating everybody else in the set simultaneously. Therefore, the number N of harmonic sets is at most

$$\binom{n}{4} - \sum_{k=1}^n \binom{d_k}{3}.$$

Note that the binomial function $\binom{x}{3}$ is convex. So we can use the majorization inequality to get

$$N \leq \binom{n}{4} - n \binom{\frac{n-3}{2}}{3} = \frac{n(n-3)(n^2+6n-31)}{48}.$$

It remains to show that this bound is attainable.

Suppose the game between player k and player $k+1$ ends in a tie for all k , where player $n+1$ means player 1. For any other game between player i and player j where $1 \leq i < j \leq n$, player j beats player i if and only if $i \equiv j \pmod{2}$.

For even k , player k beats players $2, 4, \dots, k-2$ and $k+3, k+5, \dots, n$. So player k has won $\frac{n-3}{2}$ games. For odd $k < n$, player k beats players $1, 3, \dots, k-2$ and $k+3, k+5, \dots, n-1$. So player k has won $\frac{n-3}{2}$ games. Also, player n beats players $3, 5, \dots, n-2$. So player n has won $\frac{n-3}{2}$ games. This shows all players have won the same number of games. Therefore, the equality of the majorization inequality holds.

We now show that for any set of four players in which nobody beats everyone else, it is harmonic. Once this is shown, all equalities in the above deduction hold, which means the bound is attained. Suppose on the contrary that there exists a non-harmonic set S of four players a, b, c, d such that none of them beats everyone else in this set.

- If there is no tie among the games played between a, b, c, d , then WLOG we may assume player a beats players b and c . By the assumption, player d must beat player a . Since player d does not beat everyone else, WLOG

assume player b beats player d . If player c beats player b , then we have the cycle $a \rightarrow c \rightarrow b \rightarrow d \rightarrow a$, and so the set is harmonic. If player c beats player d , WLOG assume player b beats player c . Then we have the cycle $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$, and so the set is harmonic.

The only case left is that both players b and d beat player c , which means player c is beaten by everyone else. This holds when each of a, b, d is less than and has different parity as c , or is greater than and has the same parity as c . WLOG assume $a < b < d$.

- If $a < b < d < c$, then player d beats everyone else, contradiction.
 - If $a < b < c < d$, then player b beats everyone else, contradiction.
 - If $a < c < b < d$, then player a beats everyone else, contradiction.
 - If $c < a < b < d$, then player d beats everyone else, contradiction.
- If there is a tie in the game played between c and d , then c and d are consecutive integers, or are 1 and n . By the construction, it is impossible that both c and d beat the same player, or are beaten by the same player. (This is the place where we have used n is odd, since otherwise 1 and n are beaten by some players in the same construction of the even case.)

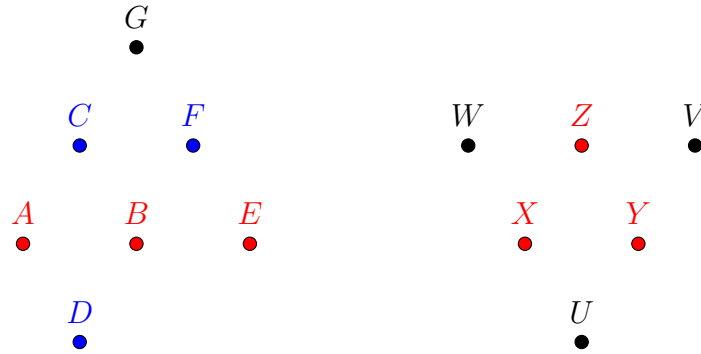
WLOG assume player a does not beat player c , and player d does not beat player a . If the same holds when a is replaced by b , then we have a cycle $c \rightarrow a \rightarrow d \rightarrow c \rightarrow b \rightarrow d \rightarrow c$, and so the set is harmonic. (Although the cycle does not consist of distinct players, we can easily shorten the sequence so that it only consists of distinct players.) If player b does not beat player d , and player c does not beat player b , then we also have a cycle $c \rightarrow a \rightarrow d \rightarrow b \rightarrow c$, and so the set is harmonic.

This shows the construction works. So the proof is complete.

Year 2014

1. We first show that there exists a monochromatic equilateral triangle of side length 2, $\sqrt{3}$ or 1.

Suppose the claim does not hold. Consider any equilateral triangle of side length 1. By the pigeonhole principle, two of the vertices are of the same colour. WLOG suppose A and B are both red, where $AB = 1$. Let C and D be points such that $\triangle ABC$ and $\triangle ABD$ are equilateral. Then both C and D are blue. Next, let E be the symmetric point of A about B . Then $\triangle CDE$ is an equilateral triangle of side length $\sqrt{3}$. Thus, E is red. Let F be the point such that both $\triangle BEF$ and $\triangle BCF$ are equilateral. As $\triangle BEF$ has side length 1, the point F is blue. Lastly, let G be the point such that both $\triangle CFG$ and $\triangle AEG$ are equilateral. No matter G is blue or red, we can find a monochromatic equilateral $\triangle CFG$ or $\triangle AEG$, which are of side length 1 and 2 respectively. Therefore, the claim is proven.



We are done unless we can only find a monochromatic equilateral $\triangle XYZ$ of side length 1. WLOG suppose X, Y, Z are red. Consider $\triangle UVW$ such that X, Y, Z are midpoints of WU, UV, VW respectively. If one of U, V, W is red, say U is red, then we obtain a monochromatic rhombus $YZXU$ of side length 1. If all of U, V, W are blue, then we obtain a monochromatic equilateral $\triangle UVW$ of side length 2. This completes the proof.

2. Let the common root be t . Adding the three equations, we get

$$(a + b + c)(t^3 + t + 1) = 0.$$

If $t^3 + t + 1 = 0$, then $at^3 + bt + c = 0$ becomes $(b - a)t + (c - a) = 0$. Similarly, $bt^3 + ct + a = 0$ becomes $(c - b)t + (a - b) = 0$. As a, b, c are distinct, we obtain $\frac{c - a}{a - b} = t = \frac{a - b}{b - c}$. After expansion and regrouping the terms, this becomes

$$(a - b)^2 + (b - c)^2 + (c - a)^2 = 0.$$

This is a contradiction since a, b, c are distinct.

Therefore, we must have $a + b + c = 0$. In that case, it is obvious that $t = 1$ is the common root. As a, b, c are nonzero, there are two possible cases.

If two of a, b, c are positive, say $a, b > 0$, we consider $f(x) = bx^3 + cx + a$. Since $f(0) = a > 0$ and $f(x) < 0$ when $x \rightarrow -\infty$, there is at least one negative root of $f(x)$. As 1 is another root and a cubic polynomial cannot have exactly two real roots, the equation $f(x) = 0$ must have exactly three real roots.

Similarly, if two of a, b, c are negative, say $a, b < 0$, then the above argument shows $g(x) = -(bx^3 + cx + a)$ has three real roots. This shows $f(x) = 0$ also has three real roots.

3. (Belarus MO 2014 Category B Problem 2 modified) The only solutions are $(a, b) = (t, t), (0, 11)$ where t is any integer.

We have

$$\begin{aligned} & (b^2 + 11(a - b))^2 = a^3b \\ \Leftrightarrow & b^4 + 22b^2(a - b) + 121(a - b)^2 = a^3b \\ \Leftrightarrow & (a - b)(121(a - b) + 22b^2 - b(a^2 + ab + b^2)) = 0. \end{aligned}$$

When $a = b$, the equation always holds. Now we may assume $a \neq b$. It remains to solve $121(a - b) + 22b^2 - b(a^2 + ab + b^2) = 0$. This means

$$ba^2 + (b^2 - 121)a + b(b - 11)^2 = 0. \quad (1)$$

If $b = 0$, we have $a = 0$, which is a contradiction. If $b \neq 0$, this is a quadratic equation in a . The discriminant is

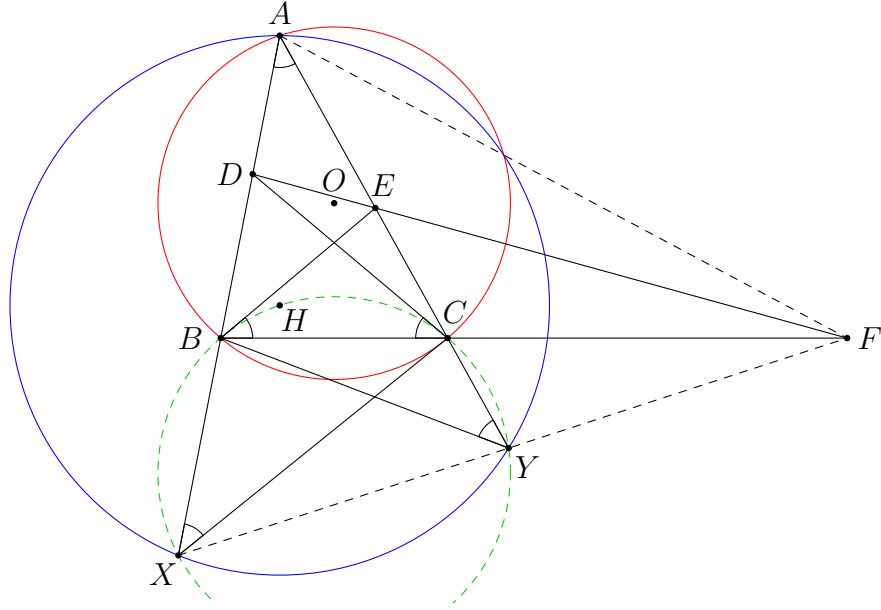
$$\Delta = (b^2 - 121)^2 - 4b^2(b - 11)^2 = -(b - 11)^3(3b + 11).$$

When $b = 11$, the equation (1) becomes $11a^2 = 0$. The only solution is $a = 0$. When $b \neq 11$, we need $-(b - 11)(3b + 11) \geq 0$. This implies $b = -3, -2, \dots, 10$. Note that Δ should be a perfect square since a is an integer. We check the value of $c = -(b - 11)(3b + 11)$ one by one as follows.

b	-3	-2	-1	1	2	3	4	5	6	7	8	9	10
c	28	65	96	140	153	160	161	156	145	128	105	76	41

None of these is a perfect square. Therefore, the only solutions are $a = b$ and $(a, b) = (0, 11)$.

4. Let O and H be the circumcentre and orthocentre of $\triangle ABC$ respectively. Let X and Y be the points on the extension of AB and AC such that $HA = HX = HY$. The centres of (ABC) and (AXY) are O and H respectively. It remains to show that F lies on the radical axis of these circles.



Firstly, since $\angle CXB = \angle CXA = \angle XAC = \angle BAY = \angle AYB = \angle CYB$, the points C, B, X, Y are concyclic. Also, since $\angle CBE = \angle BAE = \angle CXB$, the line BE is tangent to $(CBXY)$. Similarly, CD is tangent to $(CBXY)$.

Applying Pascal's theorem to the points $BBCYX$, we know that the points $BB \cap CY = E$, $CC \cap XB = D$ and $BC \cap YX$ are collinear. As $BC \cap DE = F$, this means X, Y, F are collinear. Now, the powers of F with respect to (ABC) and (AXY) are $FB \times FC$ and $FX \times FY$ respectively, which are equal since C, B, X, Y are concyclic. Therefore, F lies on the radical axis of these circles. Thus, AF is the radical axis. This implies $AF \perp OH$.

Year 2015

1. The smallest possible value of n is 42.

Firstly, it is obvious that $40 = a_1^2 + a_2^2 + \dots + a_n^2 < 1 + 1 + \dots + 1 = n$. Suppose $n = 41$. WLOG assume $a_1 \leq a_2 \leq \dots \leq a_n$ and $a_j \neq 0$ for each j . Note that $a_1 < 0 < a_{41}$ since $a_1 + a_2 + \dots + a_{41} = 0$. Let k be the index such that $a_k < 0 < a_{k+1}$. By flipping the signs of all terms if necessary, we may assume $k \leq \left\lfloor \frac{41}{2} \right\rfloor = 20$. We have

$$\begin{aligned} 40 &= a_1^2 + a_2^2 + \dots + a_{41}^2 \\ &= (a_1^2 + a_2^2 + \dots + a_k^2) + (a_{k+1}^2 + \dots + a_{41}^2) \\ &< (a_1^2 + a_2^2 + \dots + a_k^2) + (a_{k+1} + \dots + a_{41}) \\ &= (a_1^2 + a_2^2 + \dots + a_k^2) - (a_1 + a_2 + \dots + a_k) \\ &< 2k. \end{aligned}$$

This contradicts with the assumption $k \leq 20$. Therefore, $n \geq 42$. It remains to construct such a sequence when $n = 42$. Indeed, we can take $a_j = -\sqrt{\frac{20}{21}}$ for $1 \leq j \leq 21$ and $a_j = \sqrt{\frac{20}{21}}$ for $22 \leq j \leq 42$.

2. The only solution is $(x, y, z) = (4030, 4030, 28210)$ and its permutation.

Claim. If a, b, c and $\sqrt{a} + \sqrt{b} + \sqrt{c}$ are positive rational numbers, then each of $\sqrt{a}, \sqrt{b}, \sqrt{c}$ is a rational number.

Proof. Let $s = \sqrt{a} + \sqrt{b} + \sqrt{c} \in \mathbb{Q}^+$. Note that

$$\begin{aligned} &\sqrt{a} + \sqrt{b} + \sqrt{c} = s \\ \Rightarrow &\sqrt{a} + \sqrt{b} = s - \sqrt{c} \\ \Rightarrow &a + b + 2\sqrt{ab} = s^2 + c - 2s\sqrt{c} \\ \Rightarrow &2\sqrt{ab} = s^2 + c - a - b - 2s\sqrt{c}. \end{aligned}$$

Note that $t = s^2 + c - a - b \neq 0$. Therefore, this implies $4ab = t^2 + 4s^2c - 4st\sqrt{c}$, and hence $\sqrt{c} \in \mathbb{Q}$. Similarly, $\sqrt{a}, \sqrt{b} \in \mathbb{Q}$. $\square$

For simplicity, we write $n = 2015$. Using the claim, we may let $\sqrt{\frac{n}{x+y}} = \frac{a'}{a}$, $\sqrt{\frac{n}{y+z}} = \frac{b'}{b}$ and $\sqrt{\frac{n}{z+x}} = \frac{c'}{c}$ for some positive integers a', a, b', b, c', c such that $(a', a) = (b', b) = (c', c) = 1$.

Now, $\sqrt{\frac{n}{x+y}} = \frac{a'}{a}$ implies $na^2 = (a')^2(x+y)$. This yields $(a')^2 \mid n$, and so we must have $a' = 1$. Similarly, we have $b' = c' = 1$. In order that

$$\sqrt{\frac{n}{x+y}} + \sqrt{\frac{n}{y+z}} + \sqrt{\frac{n}{z+x}} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

is an integer, we must have $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1, 2, 3$ since $a, b, c \geq 1$. WLOG assume $a \leq b \leq c$.

Case 1. $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$

Note that $a \leq 3$ since $1 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{1}{a} \times 3$. Clearly, $a \neq 1$.

- If $a = 2$, then $\frac{1}{b} + \frac{1}{c} = \frac{1}{2}$, i.e. $(b-2)(c-2) = 4$. Thus, $(b, c) = (3, 6), (4, 4)$.
When $b = 3$ and $c = 6$, we have $x+y = 4n$, $y+z = 9n$ and $z+x = 36n$.
We easily obtain $x = \frac{31n}{2} \notin \mathbb{Z}$. When $b = c = 4$, we have $x+y = 4n$ and $y+z = z+x = 16n$. We find that $x = y = 2n = 4030$ and $z = 14n = 28210$.
- If $a = 3$, we must have $a = b = c = 3$. But then $x+y = y+z = z+x = 9n$ implies $x = \frac{9n}{2} \notin \mathbb{Z}$.

Case 2. $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 2$

Note that $a = 1$ as $2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{1}{a} \times 3$. This gives $\frac{1}{b} + \frac{1}{c} = 1$, i.e. $(b-1)(c-1) = 1$. The only solution is $b = c = 2$. We have $x+y = n$ and $y+z = z+x = 4n$. But then $x = \frac{n}{2} \notin \mathbb{Z}$.

Case 3. $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 3$

Clearly, we must have $a = b = c = 1$. This yields $x+y = y+z = z+x = n$. But then $x = \frac{n}{2} \notin \mathbb{Z}$.

Therefore, the only solution is $(x, y, z) = (4030, 4030, 28210)$ up to permutation.

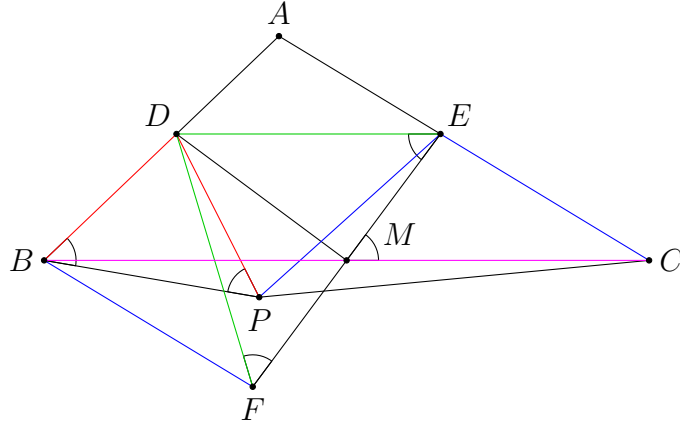
3. Note that $\angle BPD = \angle CME = \angle DEM$. So we can find a point F on the extension of EM such that $\triangle DPB \sim \triangle DEF$. By spiral similarity and since

$DB = DP$, we have $\triangle DFB \cong \triangle DEP$. As $EC = EP = FB$ and $CM = BM$, by the sine law, either $\angle FEC = \angle EFB$ or $\angle FEC + \angle EFB = 180^\circ$.

If $\angle FEC + \angle EFB = 180^\circ$, then

$$\begin{aligned}\angle AED &= 180^\circ - \angle FEC - \angle DEM \\ &= \angle EFB - \angle DEM \\ &= \angle DFB + \angle EFD - \angle DEM \\ &= \angle DEP.\end{aligned}$$

Together with $\angle AEP = 2\angle ECP$, we easily find that $DE \parallel PC$, contradicting the fact that P does not lie on BC .



Hence, we must have $\angle FEC = \angle EFB$. This yields $EC \parallel BF$. Therefore, we find that $EM = FM$. Since $DE = DF$, we have $DM \perp EF$. Now, by swapping the roles of (B, D) and (C, E) , and reversing the above arguments, we see that $\angle CPE = \angle BMD$ as desired.

4. Y has a winning strategy if and only if $n \geq 4$.

For $n = 2, 3$, after X takes away any coin, Y cannot take away any other coin. Therefore, Y does not have a winning strategy.

For $n \geq 4$, we claim that Y has a winning strategy. Since the total number $2\binom{n}{2}$ of coins is even, it suffices to partition all the coins into pairs such that the two coins in the same pair do not lie on the same circle. Once this is done, the strategy of Y is to take away the coin in the same pair as the coin just taken away by X . Clearly, Y can win as there is always a move, and the number of coins is finite.

We label the circles from 1 to n , and let $A_{ij} = A_{ji}$ and $B_{ij} = B_{ji}$ be the intersection points of circle i and circle j . For any j , we pair up $A_{j(j+1)}$ with $B_{(j+2)(j+3)}$,

where all indices are taken modulo n . Since $n \geq 4$, these two points do not lie on the same circle. Next, we pair up any other point A_{ij} with $B_{(i+1)(j+1)}$. Note that $i \not\equiv j+1 \pmod{n}$ and $j \not\equiv i+1 \pmod{n}$ by assumption. Therefore, these two points do not lie on the same circle. This gives a desired pairing, and so we are done.

Year 2016

1. The answer is $\frac{n-1}{2}$ if n is odd, $\frac{n+2}{2}$ if $n \geq 6$ is even, and 4 if $n = 4$.

Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be the sets of friends of A, B, C respectively excluding A, B, C themselves.

Case 1. n is odd

Since each of A, B, C has at least $\frac{n+1}{2}$ friends, we have $|\mathcal{A}|, |\mathcal{B}|, |\mathcal{C}| \geq \frac{n-3}{2}$. For any person X other than A, B, C , the set $\{A, B, X\}$ is a friendly group if and only if $X \in (\mathcal{A} \cap \mathcal{B})$. Similarly, $|\mathcal{B} \cap \mathcal{C}|$ and $|\mathcal{C} \cap \mathcal{A}|$ count the number of friendly groups of the form $\{B, C, X\}$ and $\{C, A, X\}$ respectively. Together with the friendly group $\{A, B, C\}$, by the inclusion-exclusion principle, the number of friendly groups is at least

$$\begin{aligned} |\mathcal{A} \cap \mathcal{B}| + |\mathcal{B} \cap \mathcal{C}| + |\mathcal{C} \cap \mathcal{A}| + 1 &= |\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{A} \cap \mathcal{B} \cap \mathcal{C}| - |\mathcal{A} \cup \mathcal{B} \cup \mathcal{C}| + 1 \\ &\geq \frac{n-3}{2} \times 3 + 0 - (n-3) + 1 = \frac{n-1}{2}. \end{aligned}$$

The minimum value $\frac{n-1}{2}$ can be attained. For example, suppose $\frac{n-3}{2}$ people are friends of A and B , and the other $\frac{n-3}{2}$ people are friends of C . Then $\{A, B, C\}$ is a friendly group and all other $\frac{n-3}{2}$ friendly groups are of the form $\{A, B, X\}$.

Case 2. n is even and $n \geq 6$

Similarly, we have $|\mathcal{A}|, |\mathcal{B}|, |\mathcal{C}| \geq \frac{n-2}{2}$. The number of friendly groups is at least

$$\begin{aligned} |\mathcal{A} \cap \mathcal{B}| + |\mathcal{B} \cap \mathcal{C}| + |\mathcal{C} \cap \mathcal{A}| + 1 &= |\mathcal{A}| + |\mathcal{B}| + |\mathcal{C}| + |\mathcal{A} \cap \mathcal{B} \cap \mathcal{C}| - |\mathcal{A} \cup \mathcal{B} \cup \mathcal{C}| + 1 \\ &\geq \frac{n-2}{2} \times 3 + 0 - (n-3) + 1 = \frac{n+2}{2}. \end{aligned}$$

The minimum value $\frac{n+2}{2}$ can be attained. For example, suppose $\frac{n-4}{2}$ people are friends of A and B , one person is a friend of B and C , one person is a friend of C and A , and the other $\frac{n-6}{2}$ people are friends of C . Then $\{A, B, C\}$ is a friendly group. In addition, there are $\frac{n-4}{2}$ friendly groups of the form $\{A, B, X\}$, one friendly group of the form $\{B, C, Y\}$, and one friendly group of the form $\{C, A, Z\}$.

Case 3. $n = 4$

Since each of A, B, C has more than 2 friends, every pair of the four people must be friends of each other. Clearly, there are 4 friendly groups.

2. The answer is $\frac{4^k - 1}{3}$ if $3 \nmid k$, and $\frac{4^k + 2}{3}$ if $3 \mid k$.

Note that n has exactly k digits, possibly starting with 0. We need to count the number N of integers of the form $\overline{a_1 a_2 \dots a_k}$ such that each a_j is one of 0, 1, 2, 7 and $3 \mid a_1 + a_2 + \dots + a_k$. Consider the generating function

$$f(x) = (1 + x + x^2 + x^7)^k = \sum_{\substack{a_1, \dots, a_k \\ a_j \in \{0, 1, 2, 7\}}} x^{a_1 + a_2 + \dots + a_k}.$$

Let c_j be the coefficient of x^j for each j . Then $N = \sum_{3 \mid j} c_j$.

Let $\omega = \frac{-1}{2} + \frac{\sqrt{3}}{2}i$ be a cube root of unity. This means a number satisfying $\omega^3 = 1$ and $\omega \neq 1$. It has the property that $\omega^{2j} + \omega^j + 1 = 0$ if $3 \nmid j$, and $\omega^{2j} + \omega^j + 1 = 3$ if $3 \mid j$. It follows that

$$\begin{aligned} 3N &= \sum_j c_j (1^j + \omega^j + \omega^{2j}) \\ &= f(1) + f(\omega) + f(\omega^2) \\ &= 4^k + (1 + \omega + \omega^2 + \omega^7)^k + (1 + \omega^2 + \omega^4 + \omega^{14})^k \\ &= 4^k + \omega^k + \omega^{2k}. \end{aligned}$$

If $3 \nmid k$, we have $\omega^k + \omega^{2k} = -1$. This gives $N = \frac{4^k - 1}{3}$. If $3 \mid k$, we have $\omega^k = \omega^{2k} = 1$. This gives $N = \frac{4^k + 2}{3}$.

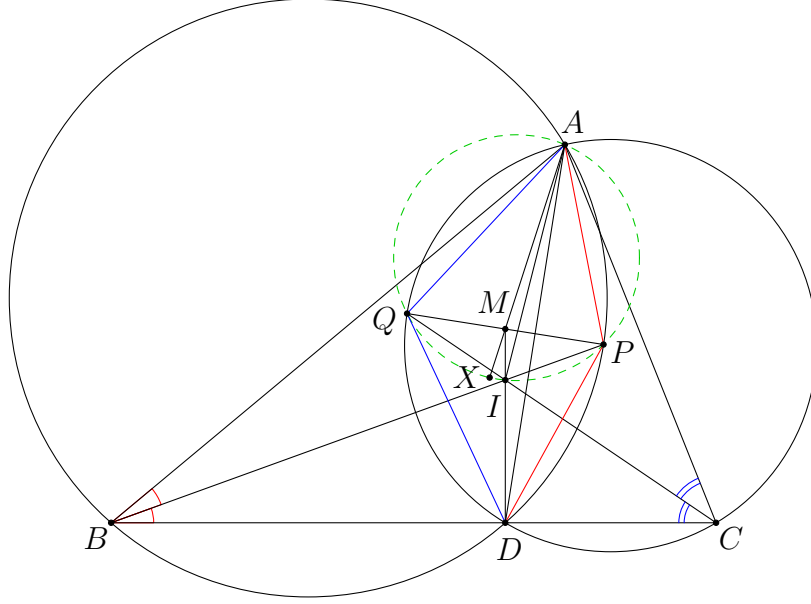
3. Since $\angle AQI = \angle AQC = \angle ADC = 180^\circ - \angle ADB = 180^\circ - \angle APB = 180^\circ - \angle API$, the points A, Q, I, P are concyclic. Let DI meet PQ at M , and let AM meet $(AQIP)$ again at X . Since $[PID] = [QID]$, we know that $MP = MQ$. Also, note that $AP = PD$ and $AQ = QD$ since they are chords opposite to some equal angles in $(ABDP)$ and $(AQDC)$. This shows PQ is the perpendicular bisector of AD . Therefore, we have

$$\angle IMP = \angle DMP = \angle AMP = \angle XMQ.$$

As M is the midpoint of PQ and Q, X, I, P are concyclic, this implies $QXIP$ is an isosceles trapezoid. Thus, $\angle QAX = \angle IAP$. Since AM is the A -median of

$\triangle AQP$, AI is the A -symmedian of $\triangle AQP$. It follows that $PI \times AQ = QI \times AP$. Finally,

$$PI \times QD = PI \times AQ = QI \times AP = QI \times PD.$$



4. (Belarus MO 2014 Category C Problem 5) The answer is $\frac{1}{2}$.

Firstly, when $\lambda = \frac{1}{2}$, we have

$$\begin{aligned} \frac{a+b}{2} &\geq \frac{1}{2}\sqrt{ab} + \frac{1}{2}\sqrt{\frac{a^2+b^2}{2}} \\ \Leftrightarrow (a+b)^2 &\geq ab + \frac{a^2+b^2}{2} + 2\sqrt{ab} \cdot \sqrt{\frac{a^2+b^2}{2}} \\ \Leftrightarrow (a+b)^2 &\geq 2\sqrt{2ab(a^2+b^2)} \\ \Leftrightarrow (a+b)^4 &\geq 8ab(a^2+b^2) \\ \Leftrightarrow (a-b)^4 &\geq 0. \end{aligned}$$

The inequality clearly holds.

Secondly, suppose the inequality holds for all $a, b > 0$. Note that

$$\begin{aligned} \frac{a+b}{2} &\geq \lambda\sqrt{ab} + (1-\lambda)\sqrt{\frac{a^2+b^2}{2}} \\ \Leftrightarrow \sqrt{2}(\sqrt{a^2+b^2} - \sqrt{2ab})\lambda &\geq \sqrt{2(a^2+b^2)} - (a+b) \\ \Leftrightarrow \frac{\sqrt{2}(a-b)^2}{\sqrt{a^2+b^2} + \sqrt{2ab}}\lambda &\geq \frac{(a-b)^2}{\sqrt{2(a^2+b^2)} + (a+b)}. \end{aligned}$$

WLOG assume $a \neq b$. Then this holds if and only if

$$\lambda \geq \frac{\sqrt{a^2 + b^2} + \sqrt{2ab}}{\sqrt{2}(\sqrt{2(a^2 + b^2)} + (a + b))}.$$

We can rewrite this as

$$2\lambda \geq 1 - \frac{a + b - 2\sqrt{ab}}{\sqrt{2(a^2 + b^2)} + (a + b)} = 1 - \frac{(\sqrt{a} - \sqrt{b})^2}{\sqrt{2(a^2 + b^2)} + (a + b)}.$$

Consider $a = 1$ and $b = 1 + \varepsilon$ where $\varepsilon \rightarrow 0$. It is clear that the right-hand side approaches 1. Therefore, in order that the inequality holds, we must have $2\lambda \geq 1$, which means $\lambda \geq \frac{1}{2}$.

Combining these, the smallest possible value of λ is $\frac{1}{2}$.

Year 2017

1. We provide two solutions.

Solution 1. The recurrence relation can be rewritten as $x_{k+1} - 3 = x_k(x_k - 3)$. By repeating the process, we obtain

$$x_{k+1} - 3 = x_k(x_k - 3) = x_k x_{k-1}(x_{k-1} - 3) = \cdots = x_k x_{k-1} \cdots x_1(x_1 - 3).$$

Changing the index, this gives $x_k = 10x_2 \cdots x_{k-1} + 3$. We now prove the assertion by induction. The base cases $k = 1$ and $k = 2$ follow from $x_1 > 3$ and $x_2 = 13 > 9$. For the inductive step, we obtain

$$x_k > 10x_2 \cdots x_{k-1} > 10 \prod_{j=2}^{k-1} 3^{2^{j-1}} = 10 \times 3^{2+2^2+2^3+\cdots+2^{k-2}} = 10 \times 3^{2^{k-1}-2} > 3^{2^{k-1}}.$$

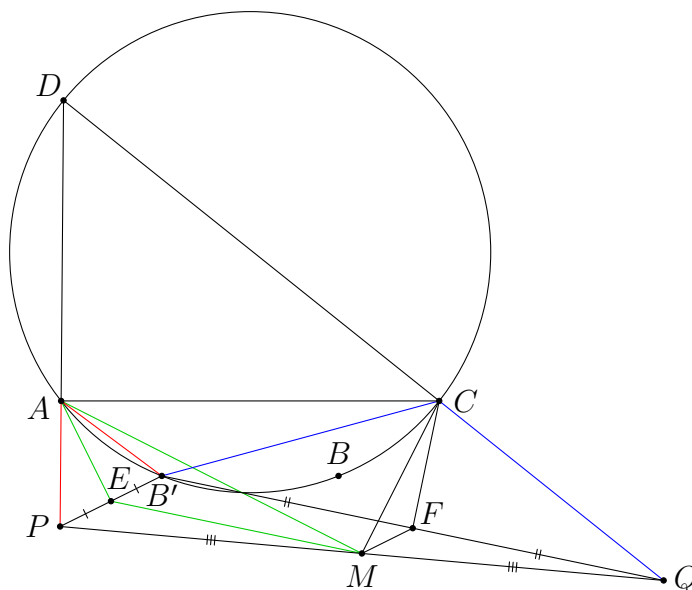
Thus, this holds for every positive integer k by induction.

Solution 2. We prove by induction that $x_k \geq 3^{2^{k-1}} + 2$ for any k . The base case holds since $x_1 = 5$. Assuming $x_k \geq 3^{2^{k-1}} + 2$ for some k , we have

$$x_{k+1} = x_k(x_k - 3) + 3 \geq (3^{2^{k-1}} + 2)(3^{2^{k-1}} - 1) + 3 > (3^{2^{k-1}})^2 + 2 = 3^{2^k} + 2.$$

The claim holds by induction. This obviously implies $x_k > 3^{2^{k-1}}$.

2. Let B' be the point such that $AB'BC$ is an isosceles trapezoid. Thus, B' lies on the circle passing through A, B, C, D . Also, we have $AP = CB = AB'$ and $CQ = AB = CB'$. Let E and F be the midpoints of $B'P$ and $B'Q$ respectively. Then $AE \perp B'P$ and $CF \perp B'Q$.



First of all, as $\angle PAB' + \angle QCB' = (180^\circ - \angle B'AD) + (180^\circ - \angle B'CD) = 180^\circ$, we know that $\angle EAB' + \angle FCB' = 90^\circ$. This implies the right-angled triangles AEB' and $B'FC$ are similar. Hence, $\frac{AE}{AB'} = \frac{B'F}{B'C}$. By the midpoint theorem, $B'EMF$ is a parallelogram. Therefore, we have

$$\frac{AE}{EM} = \frac{AE}{B'F} = \frac{AB'}{B'C}.$$

Together with

$$\begin{aligned}\angle AEM &= 90^\circ + \angle B'EM = 270^\circ - \angle EB'F \\ &= 270^\circ - (360^\circ - \angle AB'C - \angle AB'E - \angle CB'F) = \angle AB'C,\end{aligned}$$

we find that $\triangle AEM \sim \triangle AB'C$. This yields $\triangle AEB' \sim \triangle AMC$ by spiral similarity. Therefore, $\angle AMC = \angle AEB' = 90^\circ$.

3. We will prove the assertion when $\ell = k!$ by induction on k . The base case $k = 1$ is obvious.

If $k > 1$, then $\varphi(k) < k$. Suppose m and n are positive integers relatively prime to ℓ such that $m^m \equiv n^n \pmod{k!}$. This clearly implies $m^m \equiv n^n \pmod{\varphi(k)!}$. By the induction hypothesis, we know that $m \equiv n \pmod{\varphi(k)}$. Thus, there exists a positive integer a such that $am \equiv an \equiv 1 \pmod{\varphi(k)}$. Now, by the Euler-Fermat theorem, we have

$$m \equiv m^{am} \equiv n^{an} \equiv n \pmod{k}.$$

The proof is complete by induction.

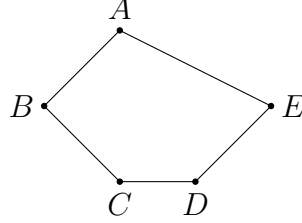
4. We shall prove the assertion for any positive integer n when 2017 is replaced by n .

Claim 1. If the points X, Y, Z lie inside or on the boundary of a polygon $\mathcal{P}$, then $[XYZ]$ does not exceed the area of one of the triangles formed by the vertices of $\mathcal{P}$. When equality holds, X, Y, Z must lie on the boundary of $\mathcal{P}$.

Proof. If X lies in the interior of $\mathcal{P}$, we can move X away from the line YZ in the direction perpendicular to YZ until X lies on the boundary of $\mathcal{P}$. The area is increased. In the same way, we may assume X, Y, Z lie on the boundary of $\mathcal{P}$. Next, suppose X lies on the side PQ where P and Q are vertices of $\mathcal{P}$. WLOG assume the distance from P to YZ is at least that from X to YZ . Then $[PYZ] \geq [XYZ]$ and we can move X to P . Similarly, we can move all three points to some vertices of $\mathcal{P}$. The area is no less than the original $[XYZ]$. $\square$

Let $\mathcal{P}$ be the convex hull of the n points. Using claim 1, all points inside $\mathcal{P}$ cannot form a good triangle. WLOG we may assume the n given points form the convex n -gon $\mathcal{P}$.

Claim 2. Let $ABCDE$ be a convex pentagon. If both $\triangle AWX$ and $\triangle AYZ$ are good triangles where $\{W, X, Y, Z\} = \{B, C, D, E\}$, then these triangles must be $\triangle ABD$ and $\triangle ACE$.

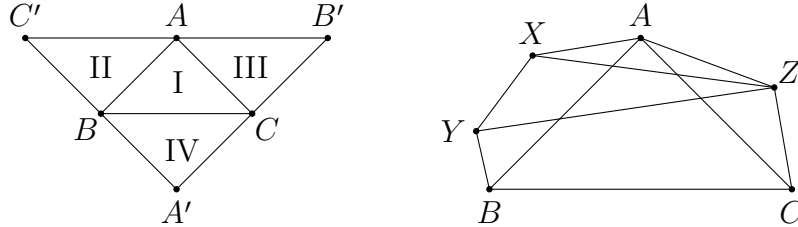


Proof. Suppose $\triangle ABC$ and $\triangle ADE$ are good triangles. As $[ABC] \geq [ABD]$, we have $\angle B + \angle C \leq 180^\circ$. Similarly, as $[ADE] \geq [ACE]$, we have $\angle D + \angle E \leq 180^\circ$. These yield $\angle A \geq 180^\circ$, which is impossible.

Suppose $\triangle ABE$ and $\triangle ACD$ are good triangles. As $[ABE] \geq [ACE]$, we have $\angle A + \angle B \leq 180^\circ$. Since $[ACD] \geq [ABD]$, we have $\angle B + \angle BAD \geq 180^\circ$. This implies $\angle B + \angle A > 180^\circ$, which contradicts the former inequality.

The only possibility is the one we claim. $\square$

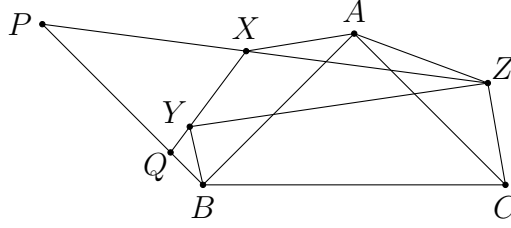
Claim 3. Suppose $\triangle ABC$ and $\triangle XYZ$ are good. Construct $\triangle A'B'C'$ such that $\triangle ABC$ is the medial triangle of $\triangle A'B'C'$. Then the points X, Y, Z must lie in the regions II, III, IV as shown and no two of them lie in the same region.



Proof. Firstly, if X lies on different sides of $B'C'$ as B , then $[XBC] > [ABC]$. Thus $\triangle ABC$ cannot be good. In the same way, all of X, Y, Z must lie in $\triangle A'B'C'$. By our assumption on the n points, none of X, Y, Z can lie inside $\triangle ABC$, which is region I. If all of X, Y, Z lie in the same region, say II, then $[XYZ] < [ABC'] = [ABC]$. This contradicts $[XYZ] = [ABC]$.

WLOG, it remains to consider the case when X, Y lie in region II and Z lies in region III. WLOG assume $AXYB$ is convex and the points lie in this order. Since $[XYZ] \geq [XBZ]$, we have $\angle ZXY + \angle XYB \leq 180^\circ$. But this cannot be true. Indeed, let ZX and XY meet the line $C'A'$ at P and Q respectively. In view of the configuration, P, Q, B must lie in that order. We find that

$$\angle ZXY + \angle XYB \geq \angle ZXQ + \angle XQB = 180^\circ + \angle ZPB > 180^\circ.$$



This shows no two of X, Y, Z can lie in the same region as desired. $\square$

Suppose the vertices of $\mathcal{P}$ are $A_1, A_2, \dots, A_n$ in that order. Suppose there are m good triangles $A_{a_i}A_{b_i}A_{c_i}$ where $a_i < b_i < c_i$. WLOG, the indices can be rearranged in dictionary order. This means $a_1 \leq a_2 \leq \dots \leq a_m$. If $a_i = a_{i+1}$, then $b_i \leq b_{i+1}$. If $a_i = a_{i+1}$ and $b_i = b_{i+1}$, then $c_i < c_{i+1}$.

Claim 4. We have

$$a_1 \leq a_2 \leq \dots \leq a_m \leq b_1 \leq b_2 \leq \dots \leq b_m \leq c_1 \leq c_2 \leq \dots \leq c_m.$$

Proof. The relation $a_1 \leq a_2 \leq \dots \leq a_m$ follows from the construction. Next, to show that $b_1 \leq b_2 \leq \dots \leq b_m$, we first assume on the contrary that $b_i > b_{i+1}$ for some i . Note that $a_i \neq a_{i+1}$ by construction. Then $A_{a_i}, A_{a_{i+1}}, A_{b_{i+1}}, A_{b_i}$ are distinct and lie in that order. No matter A_{c_i} equals $A_{c_{i+1}}$ or not, claim 2 or claim 3 is violated.

Next, to show $c_1 \leq c_2 \leq \dots \leq c_m$, we assume on the contrary that $c_i > c_{i+1}$ for some i . If $a_i = a_{i+1}$, then we must have $b_i < b_{i+1}$. This contradicts claim 2. If $a_i < a_{i+1}$, no matter A_{b_i} equals $A_{b_{i+1}}$ or not, claim 2 or claim 3 is violated.

It remains to show $a_m \leq b_1$ and $b_m \leq c_1$. For the former inequality, we first assume $a_m > b_1$. If c_1 equals one of a_m, b_m, c_m , then claim 2 is violated. If c_1 is distinct from a_m, b_m, c_m , then claim 3 is violated. Thus, we must have $a_m \leq b_1$. By symmetry (reversing the role of the two triangles), we obtain the latter inequality $b_m \leq c_1$ as well. This completes the proof of claim 4. $\square$

Let $s_i = a_i + b_i + c_i$. By claim 4, $\{s_i\}$ is an increasing sequence. Note that it is strictly increasing since the good triangles are distinct. As

$$s_m - s_1 = c_m + (b_m - c_1) + (a_m - b_1) - a_1 \leq n + 0 + 0 - 1 = n - 1,$$

the sequence contains at most n terms, meaning that at most n good triangles are possible. This completes the proof.

Year 2018

1. By the AM-GM inequality, we have

$$\begin{aligned}\frac{bc}{a} + \frac{ca}{b} &= c \left(\frac{b}{a} + \frac{a}{b} \right) \geq 2c, \\ \frac{ca}{b} + \frac{ab}{c} &= a \left(\frac{c}{b} + \frac{b}{c} \right) \geq 2a, \\ \frac{ab}{c} + \frac{bc}{a} &= b \left(\frac{a}{c} + \frac{c}{a} \right) \geq 2b.\end{aligned}$$

Adding the three inequalities, we get

$$\frac{bc}{a} + \frac{ca}{b} + \frac{ab}{c} \geq a + b + c. \quad (1)$$

In addition, we have

$$a + b + c = \sqrt{(a^2 + b^2 + c^2) + 2(ab + bc + ca)} \geq \sqrt{3(ab + bc + ca)} \geq \sqrt{3}. \quad (2)$$

Combining (1) and (2), we obtain $\frac{bc}{a} + \frac{ca}{b} + \frac{ab}{c} \geq \sqrt{3}$. This yields $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{\sqrt{3}}{abc}$.

Equality holds when $a = b = c = \frac{1}{\sqrt{3}}$.

2. (CGMO 2012 Problem 8 modified) The answer is 2146.

Note that $2188 = 4 \times 547$, where 547 is a prime. So $2188 = 40_{(547)}$ (base 547 representation). Let $k = \overline{ab}_{(547)}$. If k is not divisible by 547, then $b > 0$. By Lucas' theorem,

$$\binom{2188}{k} = \binom{40_{(547)}}{\overline{ab}_{(547)}} \equiv \binom{4}{a} \binom{0}{b} = 0 \pmod{547}.$$

This shows $\binom{2188}{k}$ is divisible by 547. If $547 \mid k$, then $b = 0$ and $a \leq 4$. By Lucas' theorem,

$$\binom{2188}{k} = \binom{40_{(547)}}{\overline{a0}_{(547)}} \equiv \binom{4}{a} \binom{0}{0} = \binom{4}{a} \not\equiv 0 \pmod{547}.$$

This shows $\binom{2188}{k}$ is not divisible by 547.

Next, let N be the highest power of 2 dividing $\binom{2188}{k}$. Using binary representation, we have $2188 = 100010001100_{(2)}$. By Kummer's theorem, N is the number of carries when k is added to $2188 - k$ in base 2.

Firstly, $\binom{2188}{k}$ is odd if and only if $N = 0$. This holds if and only if a_j is 0 whenever the corresponding digits of 2188 in binary representation are 0. In other words, $k = \overline{a000b000cd00}_{(2)}$. There are $2^4 = 16$ such numbers.

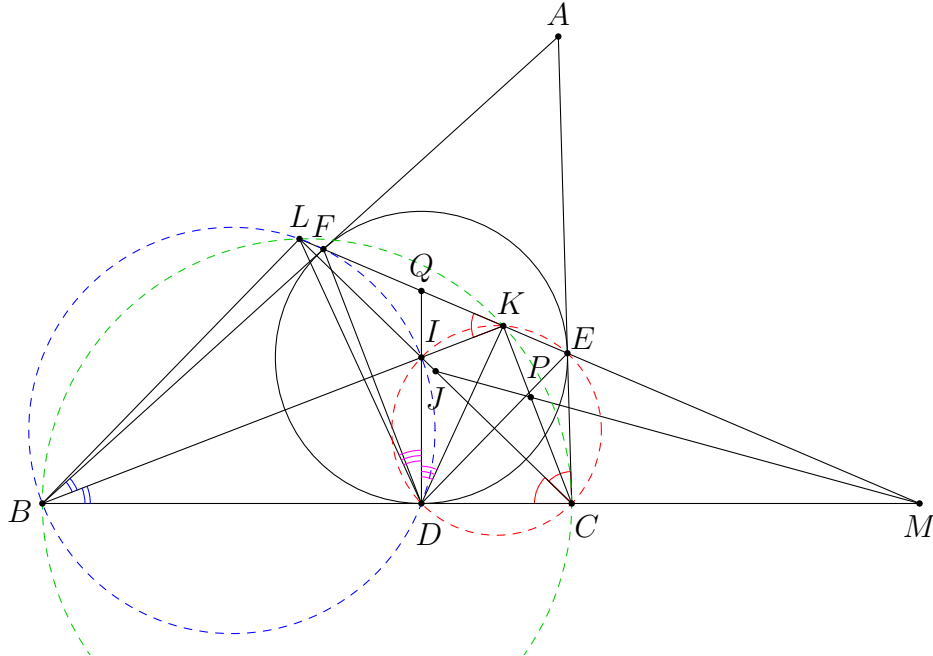
Secondly, $\binom{2188}{k}$ is even and is not divisible by 4 if and only if $N = 1$. This holds if and only if k has the form $\overline{0100a000bc00}_{(2)}$, $\overline{a0000100bc00}_{(2)}$, $\overline{a000b000c010}_{(2)}$. There are $2^3 \times 3 = 24$ such numbers.

Finally, $\binom{2188}{0} = \binom{2188}{2188} = 1$ are odd, while $4 \nmid \binom{2188}{547}, \binom{2188}{1094}, \binom{2188}{1641}$ since $547 = \overline{001000100011}_{(2)}$, $1094 = \overline{010001000110}_{(2)}$, $1641 = \overline{011001101001}_{(2)}$. Therefore, the final answer is

$$2189 - 5 - 16 - 24 + 2 = 2146.$$

3. (a) Since $\angle FKI = \angle FKB = \angle EFA - \angle KBF = 90^\circ - \frac{A}{2} - \frac{B}{2} = \frac{C}{2} = \angle ECI$, the points C, E, K, I are concyclic. Note that C, E, I, D are also concyclic. This shows C, E, K, I, D lie on the same circle. It follows that

$$\angle BKC = \angle IKC = \angle IEC = 90^\circ.$$



- (b) Let the midpoint of CL be J . Applying Menelaus' theorem using the line MPJ and $\triangle CKL$, we get $\frac{KP}{PC} \cdot \frac{CJ}{JL} \cdot \frac{LM}{MK} = 1$. Using $CJ = JL$, this simplifies

to

$$\frac{KP}{PC} = \frac{KM}{ML}. \quad (1)$$

Now, note that B, D, I, F, L are concyclic and $\angle BLC = 90^\circ$ as similar to the proof of part (a). Since $\angle BLC = \angle BKC = 90^\circ$, we know that B, C, K, L are concyclic. Hence,

$$\angle QDL = \angle IDL = \angle IBL = \angle KBL = \angle KCL = \angle KCI = \angle KDI = \angle KDQ.$$

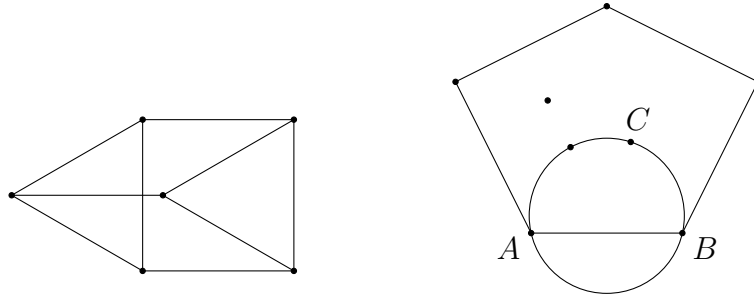
Together with $DQ \perp DM$, the lines DQ and DM are the internal and external angle bisectors of $\angle KDL$ respectively. So we get

$$\frac{KQ}{QL} = \frac{KD}{DL} = \frac{KM}{ML}. \quad (2)$$

Combining (1) and (2), we obtain $\frac{KP}{PC} = \frac{KQ}{QL}$, which yields $PQ \parallel CL$.

4. The answer is any integer $n \geq 6$.

When $n = 6$, consider two equilateral triangles of side lengths 1 having parallel sides and the same orientation such that each pair of corresponding vertices is at a distance 1 apart. It is clear that each point has a distance 1 to three other points, so that it is the circumcentre of some triangle.



Assume there are n distinct points such that each point is the circumcentre of some triangle. Consider the convex hull of these n points. Let AB be a side of the convex hull. We may assume there is no point on the segment AB . Let C be a point such that $\angle ACB$ is the largest possible. Note that $\angle ACB > 0^\circ$ since it is not possible that all n points are collinear.

If $\angle ACB \geq 90^\circ$, then the circumcentre O of $\triangle ABC$ lies on AB or on a different side to the line AB as C . In both cases, O is distinct from all these points. We can take O as the $(n+1)$ st point so that each point is the circumcentre of some triangle.

If $\angle ACB < 90^\circ$, then the circumcentre O of $\triangle ABC$ satisfies

$$\angle AOB = 2\angle ACB > \angle ACB.$$

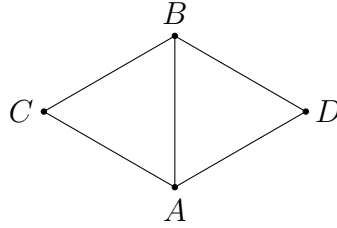
From the choice of C , the point O must be distinct from all the points. Again, we can take O as the $(n+1)$ st point.

By induction, we find that the property is satisfied for any $n \geq 6$.

It remains to show that $n = 5$ does not satisfy the property. If this is true, then both $n = 3$ and $n = 4$ do not satisfy the property from the above inductive step.

Let the points be A, B, C, D, E . WLOG assume A is the circumcentre of $\triangle BCD$. One of the following must occur.

- If B is the circumcentre of $\triangle ACD$, then $AC = AD = AB = BC = BD$. So both $\triangle ABC$ and $\triangle ABD$ are equilateral triangles. Note that at least one of A, B must lie on a circle with centre C that contains 3 points. So E lies on the circle with centre C that passes through A and B . Similarly, E lies on the circle with centre D that passes through A and B . This is a contradiction since the two circles only meet at A and B .



- If B is the circumcentre of $\triangle ACE$, then $AC = AD = AB = BC = BE$. So $\triangle ABC$ is an equilateral triangle. Thus, there must be 3 points lying on the circle with centre C that passes through A and B . In either case $CA = CB = CD$ or $CA = CB = CE$, we get the same configuration as above and hence it is a contradiction.
- If B is the circumcentre of $\triangle CDE$, we may also assume C is the circumcentre of $\triangle BDE$ since otherwise it is included in the above two cases by symmetry. Then this is exactly the same as the first case. Again, this yields a contradiction.

Therefore, $n = 5$ does not satisfy the property. The result then follows.

Year 2019

1. (USAMO 1973 Problem 2 modified) The two sequences are $a_n = 1, 10, 23, 76, \dots$ and $b_n = 1, 8, 28, 116, \dots$. Considering modulo 9, we have

$$\begin{aligned} a_n &\equiv 1, 1, 5, 4, 5, 4, \dots \pmod{9}, \\ b_n &\equiv 1, 8, 1, 8, \dots \pmod{9}. \end{aligned}$$

Since each term of the two sequences only depends on the two previous terms, we can show by induction that $a_n \equiv 4, 5 \pmod{9}$ for $n \geq 3$ and $b_n \equiv 1, 8 \pmod{9}$ for all n . Therefore, $a_m \neq b_n$ whenever $m \geq 3$.

Clearly the two sequences are strictly increasing. Thus, it is easy to see that the number 10 does not appear in the second sequence. Therefore, the only common number appearing in both sequences is $a_1 = b_1 = 1$.

2. The minimum possible score is 10.

Suppose we have a partition $S_1, S_2, \dots, S_n$ with score x .

Claim 1. If there are at least two subsets, say S_1, S_2 , both with averages at least x , then merging these two subsets into one subset will not increase the score.

Proof. Let a' be the new average of the merged set. Note that $a' \leq \max\{a_1, a_2\}$. Together with $x \leq a_1, a_2$, we obtain $a' + x \leq a_1 + a_2$. It follows that

$$\frac{a' + a_3 + a_4 + \dots + a_n}{n-1} \leq \frac{-x + a_1 + a_2 + a_3 + \dots + a_n}{n-1} = \frac{-x + nx}{n-1} = x.$$

This means the new score is at most x . □

Claim 2. If there is a subset, say S_1 , with average at most x and having at least two elements, then it is possible to split it into two subsets so that the score does not increase.

Proof. Let m be the smallest element in S_1 . We split S_1 into $\{m\}$ and $S_1 - \{m\}$. The new averages of these two sets are m and $\frac{|S_1|a_1 - m}{|S_1| - 1}$ respectively. Now, we have

$$\begin{aligned} &\frac{1}{n+1} \left(m + \frac{|S_1|a_1 - m}{|S_1| - 1} + a_2 + a_3 + \dots + a_n \right) \\ &= \frac{1}{n+1} \left(m + \frac{a_1 - m}{|S_1| - 1} + a_1 + a_2 + \dots + a_n \right) \\ &\leq \frac{m + (a_1 - m) + nx}{n+1} \leq \frac{x + nx}{n+1} = x. \end{aligned}$$

This means the new score is at most x . □

Using the above claims, we merge any two subsets with scores above average and subdivide any subset with score below average into singletons. Also, if there are subsets $\{a\}$ and $B \cup \{b\}$ for which $a \geq b$, then we can swap a and b to get the subsets $\{b\}$ and $B \cup \{a\}$ so that the average decreases, since

$$a + \frac{s+b}{|B|+1} > b + \frac{s+a}{|B|+1}$$

if $|B| \geq 1$.

Therefore, the optimal case must be of the form

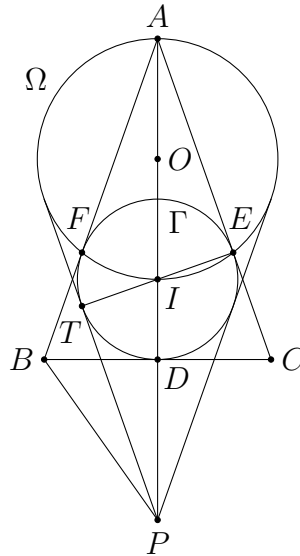
$$\{1\}, \{2\}, \{3\}, \dots, \{k\}, \{k+1, k+2, \dots, 100\}$$

for some $k \leq 100$. The score is

$$\frac{1}{k+1} \left(\frac{k(k+1)}{2} + \frac{k+101}{2} \right) = \frac{k+1}{2} + \frac{100}{2(k+1)} \geq 2\sqrt{\frac{k+1}{2} \cdot \frac{100}{2(k+1)}} = 10$$

by the AM-GM inequality. In other words, the minimum score is 10, which is attained when $k = 9$.

3. Since $\angle AFI = \angle AEI = 90^\circ$, the point I lies on Ω and the centre of Ω is the midpoint O of AI . Note that P lies on the line joining the centres of Γ and Ω . In other words, P lies on the line IO . As $AB = AC$, the line AI is simply the internal angle bisector of $\angle BAC$. In order to show that BP is perpendicular to the angle bisector BI of $\angle CBA$, it is the same as proving P is the excentre of $\triangle ABC$ opposite to the vertex A .



Let T be the point on Γ such that PT is tangent to Γ and $TP \parallel AC$. Since the tangents at T and E to Γ are parallel, TE is a diameter of Γ . This shows I is the midpoint of TE . It follows from $AE \parallel PT$ that $\triangle AIE \cong \triangle PIT$. Thus, $AI = PI$. Together with $AO = OI$, we find that $AO : OI : IP = 1 : 1 : 2$.

Note that P is the centre of homothety between Γ and Ω . Therefore, $\frac{PI}{PO}$ is the ratio of similitude, i.e. the ratio $\frac{2IE}{AI}$ of the diameters of the two circles. Thus, we have $\frac{2IE}{AI} = \frac{2}{3}$.

Let D be the midpoint of BC . As $AB = AC$, D is also the contact point of Γ with BC . From $\angle AEI = \angle ADC = 90^\circ$, we obtain $\triangle AEI \sim \triangle ADC$. Therefore, $\frac{CD}{AC} = \frac{IE}{AI} = \frac{1}{3}$. Let $AB = AC = 3k$ so that $BD = CD = k$.

Let P' be the A -excentre of $\triangle ABC$. Our goal is to show that $AI = IP'$, so that $P' = P$ as desired. Firstly, since $AD = \sqrt{(3k)^2 - k^2} = 2\sqrt{2}k$ and $\frac{AI}{ID} = \frac{AB}{BD} = 3$ by the angle bisector theorem, we have

$$AI = 2\sqrt{2}k \times \frac{3}{3+1} = \frac{3\sqrt{2}}{2}k.$$

Secondly, we have $\frac{AP'}{P'D} = \frac{AB}{BD} = 3$ by the external angle bisector theorem. It follows that

$$AP' = 2\sqrt{2}k \times \frac{3}{3-1} = 3\sqrt{2}k = 2AI.$$

Therefore, $AI = IP'$, and we are done.

4. The answer is $\frac{3}{2}$ for $n \neq 4$ and $\frac{19}{12}$ for $n = 4$.

Note that for any distinct cities A and B , exactly one of $d(A, B)$ and $d(B, A)$ is 1, while the other is at least 2. Thus it follows that $d(A, B) + d(B, A) \geq 3$. The average for this pair is at least $\frac{3}{2}$. Therefore, by considering all pairs of distinct cities, the average must be at least $\frac{3}{2}$. Equality holds if and only if $d(A, B) + d(B, A) = 3$ for all pairs of cities A and B .

Now we show that this average is attainable for $n \neq 4$ by induction.

First of all, when $n = 3$, the average is attainable if there is a road from A to B ,

B to C , C to A . For $n = 6$, consider the map given by the following table:

	A	B	C	D	E	F
A	0	1	1	1	0	0
B	0	0	1	0	1	1
C	0	0	0	1	1	1
D	0	1	0	0	0	1
E	1	0	0	1	0	0
F	1	0	0	0	1	0

(Here a ‘1’ in the AB -entry indicates a road from A to B , etc., while a ‘0’ in the AE -entry indicates no direct road from A to E , etc.). Using the table, we get the distance table:

	A	B	C	D	E	F
A	0	1	1	1	2	2
B	2	0	1	2	1	1
C	2	2	0	1	1	1
D	2	1	2	0	2	1
E	1	2	2	1	0	2
F	1	2	2	2	1	0

So $d(X, Y) + d(Y, X) = 3$ for each pair of distinct cities X and Y .

Assume when $n = k$, the average $\frac{3}{2}$ is attainable. Now, consider the case $n = k + 2$. Label the $k + 2$ cities by $A_1, A_2, \dots, A_n, X, Y$. By our assumption, there exists a way to build roads between $A_1, A_2, \dots, A_n$ such that $d(A_i, A_j) + d(A_j, A_i) = 3$ for all $1 \leq i < j \leq k$. We build a road from A_i to X for all i , a road from Y to A_i for all i , and a road from X to Y . Thus,

$$\begin{aligned}
 d(X, Y) &= 1, \\
 d(Y, X) &= 2 \quad (Y \rightarrow A_i \rightarrow X), \\
 d(A_i, X) &= 1, \\
 d(X, A_i) &= 2 \quad (X \rightarrow Y \rightarrow A_i), \\
 d(A_i, Y) &= 2 \quad (A_i \rightarrow X \rightarrow Y), \\
 d(Y, A_i) &= 1.
 \end{aligned}$$

Therefore, we see this road map produces an average distance of $\frac{3}{2}$ between any two cities.

By induction, the minimum average is attainable for all $n \geq 3$ except $n = 4$.

Now we show the average value is $\frac{19}{12}$ for $n = 4$.

We first show that the average of $\frac{3}{2}$ is unattainable. Suppose on the contrary that this average is attainable. Let the four cities be A_1, A_2, A_3, A_4 . Without loss of generality, assume there is a road from A_1 to A_2 and a road from A_1 to A_3 . Since the distance from A_2 to A_1 is 2, there must be a road from A_2 to A_4 , and a road from A_4 to A_1 . Similarly, there must be a road from A_3 to A_4 . We may assume there is a road from A_2 to A_3 . But then the distance from A_3 to A_2 is 3. This is a contradiction.

The next smallest possible average is $\frac{3 \times C_2^4 + 1}{4 \times 3} = \frac{19}{12}$. The above construction actually attains an average of $\frac{19}{12}$. Thus the minimum average of $d(A, B)$ is $\frac{19}{12}$ for $n = 4$. This completes the proof.

Year 2020

1. The greatest possible value of n is $\binom{17}{6} = 12376$.

We can identify each row as a 6-element subset of $S_{18} = \{1, 2, \dots, 18\}$, so that the elements correspond to the numbers of columns which contain 1. Then we need to maximize the number of such subsets so that no three of them form a partition of S_{18} , since forming a partition is the only case for their complement to have an empty intersection.

First we construct an example of 12376 subsets. We simply need to consider all 6-element subsets of $\{1, 2, \dots, 17\}$. Since in any three subsets, there are $6 \times 3 = 18$ elements, there must be an element belonging to at least two of the subsets by the pigeonhole principle. Therefore, the condition is satisfied. In this example, there are $\binom{17}{6} = 12376$ subsets.

Next we show that $n \leq 12376$. Suppose $S_1, S_2, \dots, S_n$ are the 6-element subsets corresponding to the rows, and let $T_1, T_2, \dots, T_{m-n}$ be the unused 6-element subsets, where $m = \binom{18}{6}$. Let N be the number of pairs (S_i, T_j) such that S_i and T_j are disjoint.

On the one hand, for each S_i , the 12 elements not in S_i can form $\frac{1}{2}\binom{12}{6}$ unordered pairs of 6-element subsets. At least one subset in each pair must be a subset T_j since no three of the S_k 's can form a partition of S_{18} . As each T_j disjoint with S_i belongs to exactly one such pair, this gives at least $\frac{1}{2}\binom{12}{6}$ pairs containing S_i . Thus, $N \geq \frac{n}{2}\binom{12}{6}$.

On the other hand, for each T_j , there are only $\binom{12}{6}$ subsets of S_{18} which are disjoint with T_j . Hence, $N \leq (m - n)\binom{12}{6}$.

Combining these, we obtain $\frac{n}{2}\binom{12}{6} \leq (m - n)\binom{12}{6}$, i.e. $n \leq \frac{2m}{3} = 12376$. This completes the proof.

2. Yes. Let $f(n)$ be the number of integers among $n + 1, n + 2, \dots, n + 2020$ having signatures less than 11. Since $2^{11} = 2048 > 2021$, all of $2, 3, \dots, 2021$ have signatures smaller than 11. Therefore, we have $f(1) = 2020$.

Next, let $p_1, p_2, \dots, p_{2020}$ be distinct primes. By the Chinese remainder theorem,

there exists a positive integer m such that

$$\begin{aligned} m &\equiv -1 \pmod{p_1^{11}}, \\ m &\equiv -2 \pmod{p_2^{11}}, \\ &\dots \\ m &\equiv -2020 \pmod{p_{2020}^{11}}. \end{aligned}$$

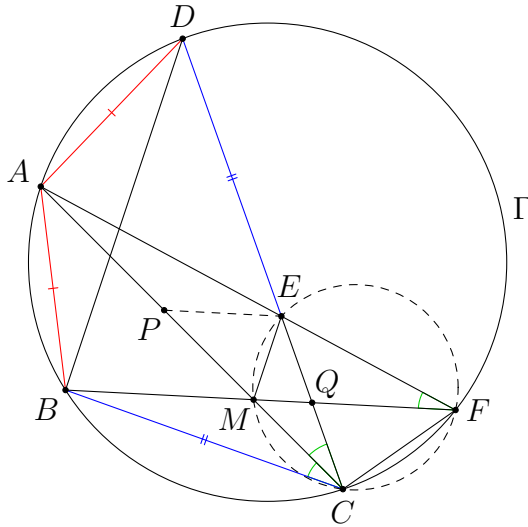
For this m , since p_j^{11} divides $m + j$ for $j = 1, 2, \dots, 2020$, each of the integers $m + 1, m + 2, \dots, m + 2020$ has signature larger than or equal to 11. This implies $f(m) = 0$.

Now, when we move from the integers $n + 1, n + 2, \dots, n + 2020$ to the integers $n + 2, n + 3, \dots, n + 2021$, the number of integers with signature less than 11 will be increased or decreased by 1, or remain unchanged. Thus, the difference between $f(n)$ and $f(n + 1)$ is at most 1. Therefore, in order to decrease $f(1) = 2020$ to $f(m) = 0$, we must come across a positive integer k such that $f(k) = 1812$, and we are done.

3. Since $\angle EFM = \angle AFB = \angle DCA = \angle ECM$, the points M, C, F, E are concyclic. Using the concyclic points, we obtain

$$\angle QME = \angle FME = \angle FCE = \angle FCD = \angle FBD = \angle QBD.$$

This implies $EM \parallel DB$, and hence $\frac{DE}{EQ} = \frac{BM}{MQ}$. Also, as $\angle MCQ = \angle MCB$, we have $\frac{BM}{MQ} = \frac{BC}{CQ}$. Combining these, we obtain $\frac{DE}{EQ} = \frac{BC}{CQ}$. It follows that $EQ = CQ$. Now, as M and Q are midpoints of CP and CE , we know that $PE \parallel MQ$ as desired.



4. By the AM-GM inequality, we have

$$\begin{aligned}a^3 + b^2 + b &\geq 3\sqrt[3]{a^3b^3} = 3ab, \\b^2 + b + 1 &\geq 3\sqrt[3]{b^3} = 3b.\end{aligned}$$

This gives $a^3 + 2b^2 + 2b + 4 \geq 3ab + 3b + 3 = 3(ab + b + 1)$. Since $abc = 1$, we can let $a = \frac{x}{y}$, $b = \frac{y}{z}$, $c = \frac{z}{x}$ for some positive real numbers x, y, z . Then we have

$$\frac{1}{a^3 + 2b^2 + 2b + 4} \leq \frac{1}{3(ab + b + 1)} = \frac{z}{3(x + y + z)}. \quad (1)$$

By symmetry, we have

$$\frac{1}{b^3 + 2c^2 + 2c + 4} \leq \frac{x}{3(y + z + x)}, \quad (2)$$

$$\frac{1}{c^3 + 2a^2 + 2a + 4} \leq \frac{y}{3(z + x + y)}. \quad (3)$$

Adding (1), (2) and (3), we obtain the desired inequality. Equality holds when $a = b = c = 1$.

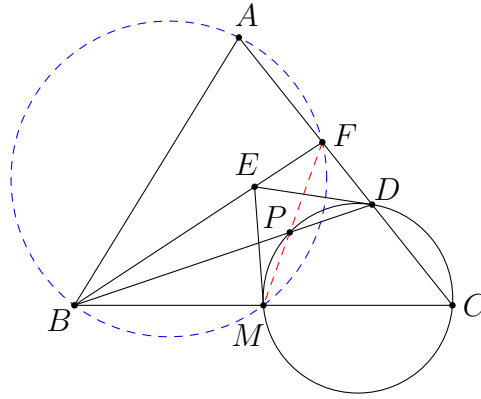
Year 2021

1. Yes. We claim that $P(x, y) = (y - 3x)(y - 3x - 1)(y - 3x - 2)$ satisfies the conditions. Clearly, it is a non-constant polynomial. For any real number α , let $n = \lfloor \alpha \rfloor$. Then we have $n \leq \alpha < n + 1$ and hence $3n \leq 3\alpha < 3n + 3$. This shows $\lfloor 3\alpha \rfloor = 3n, 3n + 1, 3n + 2$. Therefore, one of $\lfloor 3\alpha \rfloor - 3\lfloor \alpha \rfloor, \lfloor 3\alpha \rfloor - 3\lfloor \alpha \rfloor - 1$ and $\lfloor 3\alpha \rfloor - 3\lfloor \alpha \rfloor - 2$ must be 0. This shows $P(\lfloor \alpha \rfloor, \lfloor 3\alpha \rfloor) = 0$ for all $\alpha \in \mathbb{R}$.
2. We provide two solutions.

Solution 1. Suppose BD meets (CDM) again at P . Applying Pascal's theorem to the points $MMCDDP$, we know that $MM \cap DD = E$, $MC \cap DP = B$ and $CD \cap PM$ are collinear. Thus, MP passes through $BE \cap CD = F$. Now, since

$$\angle CMF = \angle ADB = \angle BAF,$$

the points A, B, M, F are concyclic.

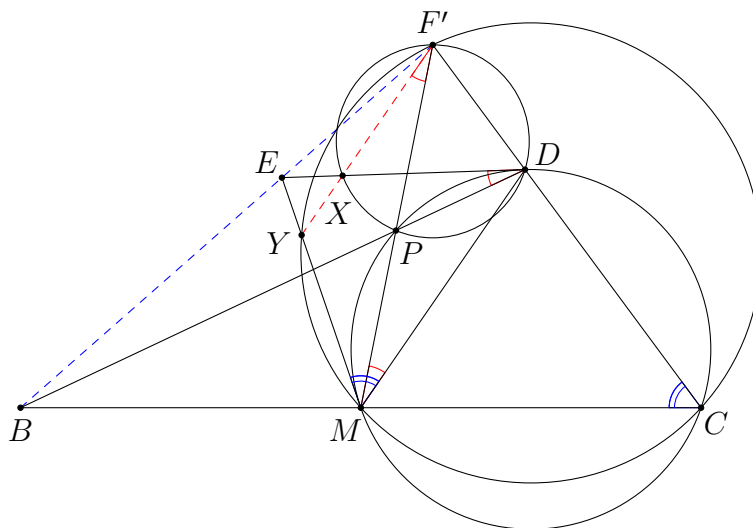


Solution 2. We provide another proof without using Pascal's theorem as follows. Again, the main task is to prove M, P, F are collinear. We define F' as the intersection point of CD and MP . It suffices to prove B, E, F' are collinear. Suppose DE meets (PDF') again at X , and ME meets (MCF') again at Y .

Firstly, since $BP \times BD = BM \times BC$, the powers of B with respect to (PDF') and (MCF') are the same. Thus, B lies on the radical axis of these circles.

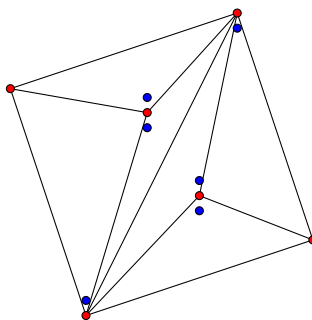
Secondly, as $\angle XF'P = \angle XDP = \angle DMP = \angle DMF'$, we have $XF' \parallel MD$. Also, as $\angle MYF' = 180^\circ - \angle MCF' = 180^\circ - \angle YMD$, we have $YF' \parallel MD$. Therefore, Y, X, F' are collinear, and $YX \parallel MD$. Since $EM = ED$, we have $EY = EX$. From this, it follows that $EX \times ED = EY \times EM$, and hence E lies on the radical axis of (PDF') and (MCF') .

Thirdly, it is clear that F' lies on the radical axis of (PDF') and (MCF') . It follows that B, E, F' are collinear as desired.



3. The answer is 4044.

After Alice has drawn the 4 red vertices of a square, we can draw one of the diagonals of the square to form 2 triangles. Each time when Alice draws one more red point inside the square, it must lie inside one of the triangles formed. By joining the new red point to the 3 vertices of this triangle, we obtain a new triangulation with 2 more triangles. At the end, when Alice has drawn all the red points, there is a triangulation with $2 + 2 \times 2021 = 4044$ triangles. Since there is a blue point in each of these triangles, we must have $n \geq 4044$.



Next, we provide a strategy for Bob so that it is sufficient to draw 4044 blue points. We set up a coordinate system such that all red points have different x -coordinates. For each red point $P(x, y)$ inside the square, Bob draws two blue points $(x, y + \varepsilon)$ and $(x, y - \varepsilon)$ where 2ε is the distance from P to the closest line joining two red points different from P . We say that these two blue points are associated to the point P . For the 4 red vertices of the square, Bob does the same thing except that points outside the square need not be drawn. Note that

there are only 2 blue points each of which is associated to a vertex of the square whose x -coordinate is neither the smallest nor the largest. In total, there are $2 \times 2021 + 2 = 4044$ blue points. We claim that the condition is satisfied.

Indeed, consider any triangle ABC with red vertices where the x -coordinates of A, B and C are in ascending order. WLOG assume B lies below the line AC . Then $\triangle ABC$ contains the blue point associated to B lying above B since this blue point also lies below the line AC . This completes the proof.

4. We shall prove that $f(n) = F_{2n+1}$, the $(2n+1)$ th Fibonacci number. Let $\zeta = e^{\frac{2\pi i}{2n+1}}$ be a primitive $(2n+1)$ th root of unity. Then

$$1 + 4 \cos^2 \left(\frac{k\pi}{2n+1} \right) = 3 + 2 \cos \left(\frac{2k\pi}{2n+1} \right) = 3 + 2 \operatorname{Re}(\zeta^k) = 3 + \zeta^k + \zeta^{-k}.$$

Note that

$$\begin{aligned} x^{2n+1} - 1 &= \prod_{k=1}^{2n+1} (x - \zeta^k) \\ &= (x - 1) \prod_{k=1}^n (x - \zeta^k)(x - \zeta^{-k}) \\ &= (x - 1) \prod_{k=1}^n (x^2 - (\zeta^k + \zeta^{-k})x + 1). \end{aligned}$$

Let $\alpha = \frac{3 + \sqrt{5}}{2}$ so that $\alpha^2 - 3\alpha + 1 = 0$. Putting $x = -\alpha$, we obtain

$$-\alpha^{2n+1} - 1 = (-\alpha - 1) \prod_{k=1}^n (\alpha^2 + (\zeta^k + \zeta^{-k})\alpha + 1) = -(\alpha + 1) \alpha^n \prod_{k=1}^n (3 + \zeta^k + \zeta^{-k}).$$

It follows that

$$f(n) = \prod_{k=1}^n (3 + \zeta^k + \zeta^{-k}) = \frac{\alpha^{2n+1} + 1}{\alpha^n(\alpha + 1)}.$$

Next, the terms of the Fibonacci sequence satisfy

$$F_{2n+5} = F_{2n+4} + F_{2n+3} = 2F_{2n+3} + F_{2n+2} = 3F_{2n+3} - F_{2n+1}.$$

Letting $a_n = F_{2n+1}$, we have the recurrence relation $a_{n+2} = 3a_{n+1} - a_n$ and the conditions $a_0 = 1$ and $a_1 = 2$. Since the roots to $\lambda^2 - 3\lambda + 1 = 0$ are α and $\frac{1}{\alpha}$, the general term is $a_n = A\alpha^n + B \cdot \frac{1}{\alpha^n}$ for some constants A and B . Using $a_0 = 1$ and $a_1 = 2$, it is easy to deduce

$$a_n = \frac{\alpha}{\alpha + 1} \cdot \alpha^n + \frac{1}{\alpha + 1} \cdot \frac{1}{\alpha^n} = \frac{\alpha^{2n+1} + 1}{\alpha^n(\alpha + 1)} = f(n).$$

Therefore, $f(n) = F_{2n+1}$ is an integer for all n .

Year 2022

1. Suppose on the contrary that all three numbers $2a - \frac{1}{b}$, $2b - \frac{1}{c}$ and $2c - \frac{1}{a}$ are greater than 2. Since $abc = 2$, at least one of a, b, c is positive. WLOG assume $a > 0$. Then $2c - \frac{1}{a} > 2$ implies $c > 1 + \frac{1}{2a} > 0$, and $2b - \frac{1}{c} > 2$ implies $b > 1 + \frac{1}{2c} > 0$.

Next, from $2b - \frac{1}{c} > 2$, we have

$$b > 1 + \frac{1}{2c}. \quad (1)$$

From $2a - \frac{1}{b} > 2$ and $abc = 2$, we have $\frac{4}{bc} - \frac{1}{b} > 2$, and so

$$b < \frac{2}{c} - \frac{1}{2}. \quad (2)$$

Combining (1) and (2), we have $1 + \frac{1}{2c} < b < \frac{2}{c} - \frac{1}{2}$, which easily implies $c < 1$. Similarly we can prove that $a < 1$ and $b < 1$. Together with the fact that they are all positive, we have $abc < 1$, contradicting the condition $abc = 2$. Therefore, at most two of $2a - \frac{1}{b}$, $2b - \frac{1}{c}$ and $2c - \frac{1}{a}$ are greater than 2.

2. Note that $\frac{39}{1428} = \frac{13}{476}$ in lowest term. Let

$$\frac{13}{476} = 0.b_1b_2\cdots b_s\overline{a_1a_2\cdots a_k} \quad (1)$$

in binary decimal representation, where $a_1a_2\cdots a_k$ is its repetend and k is its period. We can write $0.b_1b_2\cdots b_s = \frac{B}{2^s}$ and $0.a_1a_2\cdots a_k = \frac{A}{2^k}$ for some integers A and B . Thus,

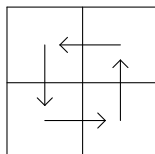
$$\frac{13}{476} = \frac{B}{2^s} + \frac{A}{2^s \cdot 2^k} + \frac{A}{2^s \cdot 2^{2k}} + \frac{A}{2^s \cdot 2^{3k}} + \cdots = \frac{B}{2^s} + \frac{A}{2^s(2^k - 1)} = \frac{B(2^k - 1) + A}{2^s(2^k - 1)}. \quad (2)$$

Note that (2) is equivalent to (1). Therefore, it remains to find the smallest positive integer k such that (2) holds for some A, B, s satisfying $0 \leq A < 2^k$ and $0 \leq B < 2^s$.

As $476 = 2^2 \times 7 \times 17$, we must have $7 \mid 2^k - 1$ and $17 \mid 2^k - 1$. Since the orders of 2 modulo 7 and 17 are 3 and 8 respectively, we have $3 \mid k$ and $8 \mid k$. Therefore, the smallest possible k is 24. Indeed, we can take $s = 2$, $B = 0$ and $A = 13 \times \frac{2^{24} - 1}{119}$, so that (2) is satisfied. This shows the period is 24.

3. The answer is $2022^2 + 1 = 4088485$.

First we show that 2022^2 is insufficient. Suppose there are 2022^2 beetles in the top left 2022×2022 square grid. This square grid can be partitioned into many 2×2 square grids in the natural way. If the beetles in each 2×2 square grid move in the anticlockwise direction as shown, then no beetle will ever meet.



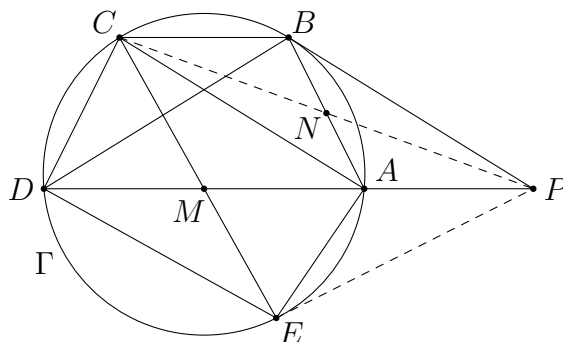
Next, suppose there are at least $2022^2 + 1$ beetles. We label all the unit squares as follows. In the odd rows, we label the unit squares $ABABAB \dots ABA$. In the even rows, we label the unit squares $CDCDCD \dots CDC$. Note that there are 1011^2 unit squares with label D .

As there are at least $2022^2 + 1$ beetles in total, we can always find at least $1011^2 + 1$ beetles lying in unit squares with the same label by the pigeonhole principle. In each move, either all beetles move in the order $A \rightarrow B \rightarrow D \rightarrow C \rightarrow A$ or all beetles move in the order $A \rightarrow C \rightarrow D \rightarrow B \rightarrow A$. In any case, these $1011^2 + 1$ beetles must move to unit squares with label D simultaneously in a certain move. As there are only 1011^2 unit squares with label D , two of the beetles must meet at a certain unit square. This completes the proof.

4. Let P be the intersection of the tangent at B to Γ and the line AD . Then we have

$$\angle ABP = \angle ACB = \angle BAC.$$

This shows $BP \parallel CA$. Note that $CB \parallel AP$ since $ABCD$ is an isosceles trapezoid. Therefore, $PBCA$ is a parallelogram. As N is the midpoint of AB , the diagonal CP passes through N .



Next, the areas of $\triangle CDE$ and $\triangle CAE$ are the same since $MD = MA$. This implies

$$\frac{1}{2}DC \times DE \sin \angle CDE = \frac{1}{2}AC \times AE \sin \angle CAE.$$

As $\angle CDE + \angle CAE = 180^\circ$, we have $\sin \angle CDE = \sin \angle CAE$. Therefore,

$$\frac{ED}{EA} = \frac{CA}{CD} = \frac{BD}{BA}.$$

This shows $ABDE$ is a harmonic quadrilateral, and hence the tangent at E to Γ passes through P . This proves the tangent at E , the line AD and the line CN are concurrent at P .